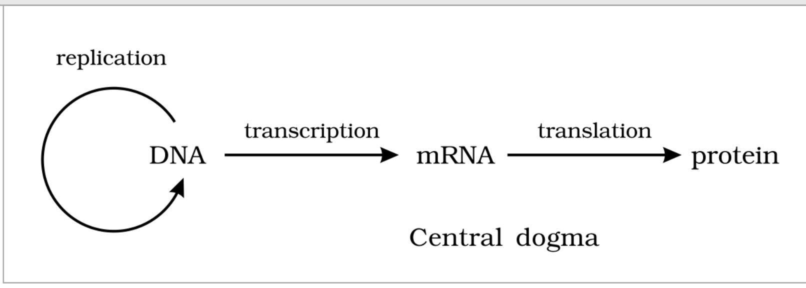
MOLECULAR BASIS OF INHERITANCE

In the previous chapter you learnt about inheritance as a pattern: Mendel's **factors**, later called **genes**, passed from parent to offspring in predictable ratios. That account never said what a gene *is* made of. This chapter answers that question at the level of the molecule. It establishes that **DNA** is the genetic material, works out how DNA is built and packaged, how it is copied (**replication**), how its information is read out into RNA (**transcription**), and -- in the second half of the chapter -- how that RNA is translated into protein and how gene expression is regulated.

The logic of the chapter is a chain of questions, each answered experimentally: What molecule carries heredity? How is it structured? How does it copy itself faithfully? How is its message expressed? Keep that chain in view -- NEET questions very often test *which experiment proved which link*, not just the end result.

> ⓘ ***[NOTE] The central dogma.*** *Francis Crick proposed the flow of genetic information in a cell as DNA to RNA to protein: DNA makes RNA (transcription), and RNA makes protein (translation). This single arrow-chain is the skeleton on which the whole chapter hangs, so it is worth fixing before anything else.*



The central dogma of molecular biology: DNA is transcribed to RNA, and RNA is translated to protein. (DNA to RNA to Protein; transcription; translation.)

5.1 THE DNA

- **DNA** (deoxyribonucleic acid) is a **long polymer of deoxyribonucleotides**. This is the definition NCERT gives for DNA, and the whole of 5.1 unpacks it.

The **length** of a DNA molecule is normally quoted as the **number of nucleotides** (or of nucleotide pairs, called **base pairs**, bp) present in it. That number is also what defines the size of an organism's genome, so the figures below are characteristic of the organism.

Organism / genome	Length of DNA	What to notice
Bacteriophage **phi x 174**	**5386** nucleotides (single-stranded)	The smallest of the standard examples; quoted in *nucleotides*, not base pairs, because the DNA is single-stranded.
Bacteriophage **lambda**	**48502** base pairs (bp)	Double-stranded, so quoted in base pairs.
Escherichia coli	**4.6×10^6** base pairs	A typical bacterial genome -- about a thousand times the lambda phage.
Haploid human genome	**3.3×10^9** base pairs	About 700 times the E. coli genome. This is the number to remember for humans.

> ☆ ***[MEMORY AID - not in NCERT]*** *The four standard genome sizes climb in rough powers of ten: phi x 174 about 5 thousand, lambda about 48 thousand, E. coli about 4.6 million, human (haploid) about 3.3 billion. Remember the sequence* ***5386 - 48502 - 4.6 million - 3.3 billion*** *as one ladder rather than four unrelated numbers.*

5.1.1 Structure of Polynucleotide Chain

Let us recapitulate the chemical structure of a polynucleotide chain, whether DNA or RNA. A nucleotide is built up in three stages, and each stage adds one component.

1. **Nitrogenous base.** Start with a nitrogen-containing ring compound. There are two classes: **purines** and **pyrimidines**.

2 **Nucleoside = base + sugar.** A nitrogenous base is linked to the **pentose sugar** through an **N-glycosidic linkage**, forming a **nucleoside** -- for example adenosine or deoxyadenosine, guanosine or deoxyguanosine, thymidine, cytidine and uridine.

3 **Nucleotide = nucleoside + phosphate.** When a **phosphate group** is linked to the 5'-OH of a nucleoside through a **phosphoester linkage**, a corresponding **nucleotide** is formed (or deoxynucleotide, depending on the sugar present).

The bases fall into two structural classes, and this distinction matters for the geometry of the helix:

Class	Members	Ring structure
Purines	**Adenine (A)** and **Guanine (G)**	Double-ring (fused) bases -- physically the larger of the two classes.
Pyrimidines	**Cytosine (C)**, and **Thymine (T)** in DNA or **Uracil (U)** in RNA	Single-ring bases -- physically the smaller class.

*[NOTE] **Cytosine is common to both DNA and RNA.** The base that differs is the second pyrimidine: **thymine is present in DNA**, while **uracil is present in RNA** in its place. Adenine and guanine, the two purines, are common to both.*

The **sugar** is the other point of difference between the two nucleic acids. Two kinds of pentose sugar are found: **ribose**, present in RNA, and **2'-deoxyribose**, present in DNA. Deoxyribose is ribose with the hydroxyl group at the 2' position replaced by hydrogen -- which is exactly what the prefix *deoxy* records.

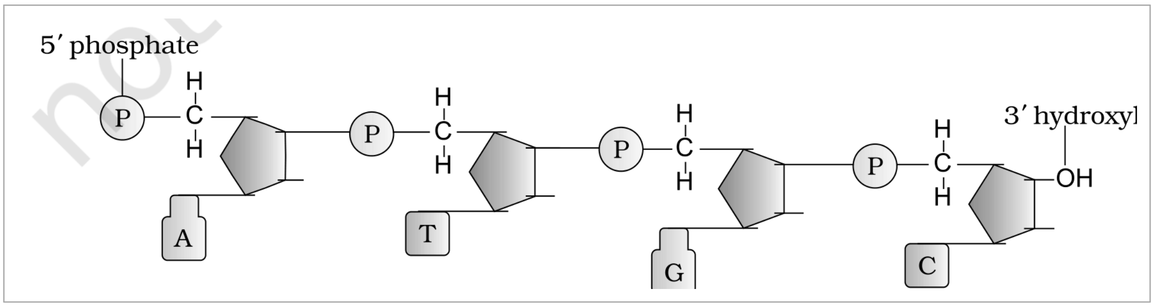


Figure 5.1 Polynucleotide chain. The 5' end carries a phosphate; the chain runs to the 3' end, which carries a free hydroxyl. Successive sugars are joined by 3'-5' phosphodiester linkages, and each sugar bears a nitrogenous base (the 5' Phosphate, Nitrogenous base, and 3' OH end are marked).

Two nucleotides are linked through a **3'-5' phosphodiester linkage** to form a **dinucleotide**. More than two nucleotides joined this way give a **polynucleotide chain**. Because the linkage always runs from the 3' position of one sugar to the 5' position of the next, the chain has a direction, and its two ends are chemically different:

End of the chain	What is free there
5' end	A free **phosphate** group at the 5'-OH of the sugar. This is the 5'-end of the polynucleotide chain.
3' end	A free **3'-OH** group (a hydroxyl). This is the 3'-end of the polynucleotide chain.

Stripping the bases off a polynucleotide leaves a repeating **sugar-phosphate** backbone, and the nitrogenous bases project from that backbone. In RNA, every nucleotide residue carries an **additional -OH group at the 2' position** of the ribose. RNA also uses uracil in place of thymine -- and note that uracil is simply **5-methyl uracil** seen from the other direction: thymine *is* 5-methyl uracil.

Who found DNA, and when. DNA as an **acidic substance present in the nucleus** was first identified by **Friedrich Meischer in 1869**, and he named it **'Nuclein'**. Its structure, however, stayed out of reach for a very long time, because a polymer that long could not be isolated intact with the techniques then available.

How the double helix was established. In 1953, **James Watson** and **Francis Crick** proposed a strikingly simple but famous **Double Helix** model for the structure of DNA. The model rested on two pieces of evidence supplied by others:

1. **Base equivalence (Erwin Chargaff).** Chargaff's observation that, for a double-stranded DNA, the **ratios between adenine and thymine, and between guanine and cytosine, are constant and equal one**.
2. **X-ray diffraction data.** The X-ray diffraction data produced by **Maurice Wilkins** and **Rosalind Franklin**, which gave the physical dimensions the model had to satisfy.

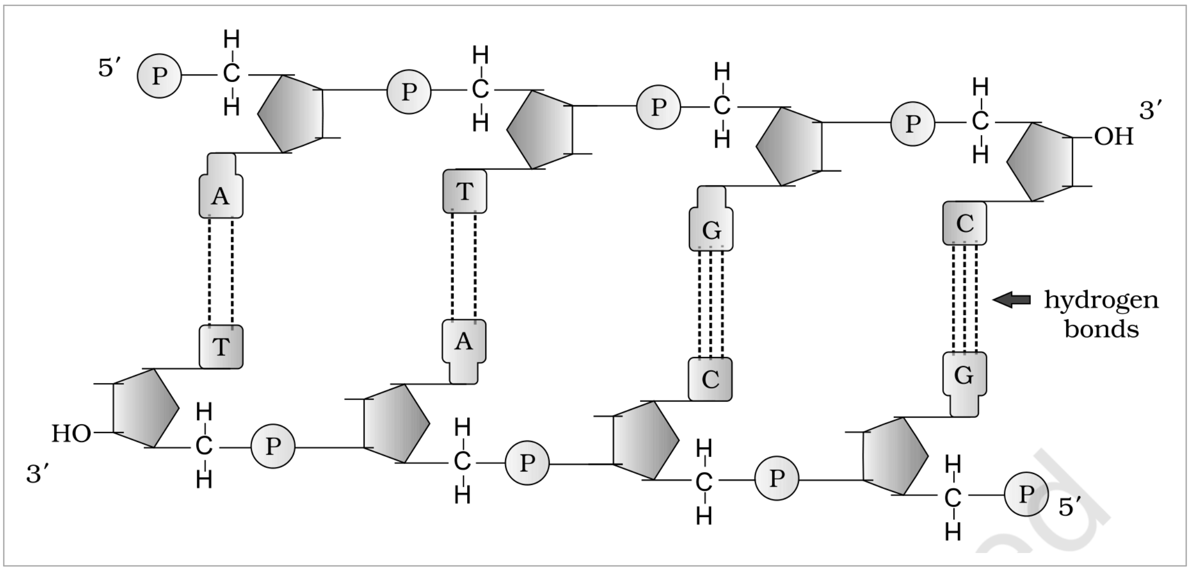


Figure 5.2 A double-stranded polynucleotide chain drawn flat. The two strands run with opposite polarity -- 5' to 3' on the upper strand and 3' to 5' on the lower -- and the base pairs A with T and G with C are held together by hydrogen bonds (the hydrogen bonds are marked; note two between A and T, three between G and C).

> ⓘ ***[NOTE] Working out one base percentage from another.*** *The exercises ask: if a double stranded DNA has 20 per cent of cytosine, calculate the per cent of adenine. Two facts already stated above are all you need. Because* ***C pairs only with G****, %C = %G, so* ***%G is also 20****. Together C and G therefore account for* ***40 per cent*** *of the bases. The DNA has* ***only four bases****, so their percentages must* ***add up to 100****, leaving* ***60 per cent*** *to be shared by A and T. Because* ***A pairs only with T****, %A = %T, so each is half of 60:* ***%A = 30 per cent*** *(and %T = 30 per cent). The general form worth remembering is* ***%A = %T****,* ***%G = %C****, and* ***%A + %T + %G + %C = 100****.*

The salient features of the Double-helix structure of DNA are these:

- It is made of **two polynucleotide chains**, where the **backbone is constituted by sugar-phosphate**, and the **bases project inside**.
- The two chains have **anti-parallel polarity**. If one chain has the polarity 5' to 3', the other has 3' to 5'.
- The bases in the two strands are paired through **hydrogen bonds (H-bonds)**, forming **base pairs (bp)**. Adenine forms **two hydrogen bonds** with thymine from the opposite strand, and **vice versa**. Guanine is bonded with cytosine by **three H-bonds**. As a result, **always a purine comes opposite to a pyrimidine**. This generates **approximately uniform distance between the two strands of the helix**.
- The two chains are **coiled in a right-handed fashion**. The **pitch of the helix is 3.4 nm** (a nanometre is one billionth of a metre, that is 10^{-9} m) and there are roughly **10 bp in each turn**. Consequently, the **distance between a base pair in a helix is approximately 0.34 nm**.
- The **plane of one base pair stacks over the other** in the double helix. This, in addition to H-bonds, confers **stability of the helical structure**.

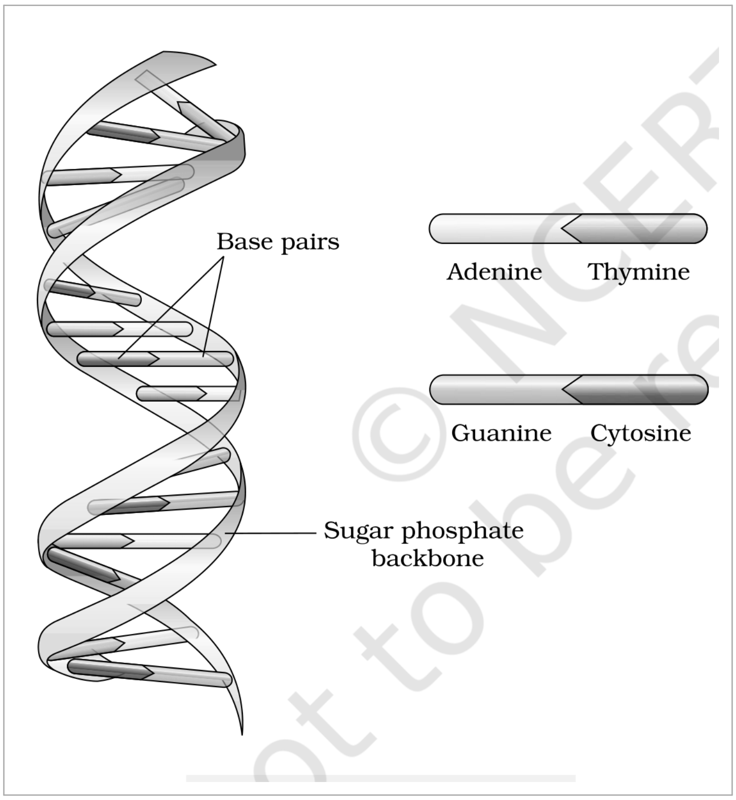


*Figure 5.3 DNA double helix. The two chains are coiled in a right-handed fashion; the **Base pairs** lie stacked inside the helix and the **Sugar phosphate backbone** runs outside. The pairing legend shows **Adenine** with **Thymine** and **Guanine** with **Cytosine**.*

> ⓘ ***[NOTE] Compare the ring structures, then read the geometry off them.*** *NCERT asks: compare the structure of purines and pyrimidines, and find out why the distance between two polynucleotide chains in DNA remains almost constant. Purines are double-ring (large) and pyrimidines single-ring (small). Since base pairing always puts a purine opposite a pyrimidine (A with T, G with C), every rung of the ladder is one large plus one small base -- so every rung spans the same width, and the two backbones stay an almost constant distance apart. A purine-purine rung would bulge and a pyrimidine-pyrimidine rung would pinch.*

> ☆ ***[MEMORY AID - not in NCERT]*** *Two numbers and two bond-counts carry most of the marks here: pitch **3.4 nm** with **10 bp per turn**, hence **0.34 nm** per base pair; and **A=T is two H-bonds, G=C is three**. The stronger G-C pair (three bonds) is why GC-rich DNA is harder to melt apart.*

5.1.2 Packaging of DNA Helix

Take the distance between two consecutive base pairs as **0.34 nm** (0.34×10^{-9} m). Then the length of the DNA double helix in a typical mammalian cell -- with its **6.6×10^{9} bp** (the diploid figure, twice the haploid 3.3×10^{9}) -- comes to **6.6×10^{9} bp x 0.34×10^{-9} m/bp = 2.2 metres**. A length of DNA far greater than the dimension of a typical nucleus, which is approximately **10^{-6} m**. How is such a long polymer packaged in a cell?

> ⓘ ***[NOTE] Working the E. coli figure the other way.*** *NCERT asks: if the length of E. coli DNA is 1.36 mm, calculate the number of base pairs. Divide length by the 0.34 nm rise per base pair: 1.36×10^{-3} m divided by 0.34×10^{-9} m/bp = **4×10^{6} bp**, which agrees with the 4.6×10^{6} bp genome quoted for E. coli in 5.1. Length and base-pair count are interconvertible through the single constant 0.34 nm per bp.*

In prokaryotes, such as E. coli, though they do not have a defined nucleus, the DNA is not scattered throughout the cell. It is held (as a large loop) in a region termed as **nucleoid**, and the DNA in the nucleoid is organised in large loops held by proteins.

In eukaryotes the packaging is more elaborate, and it begins with a set of positively charged basic proteins:

1. **Histones carry positive charge.** Histones are rich in the basic amino acid residues **lysine** and **arginine**, both of which carry positive charges in their side chains. DNA, being an acid, is negatively charged -- so the two bind electrostatically.

2 **Histone octamer.** Histones are organised to form a unit of eight molecules called a **histone octamer**.

3 **Nucleosome.** The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called a **nucleosome**. A typical nucleosome contains **200 bp of DNA helix**.

4 **Chromatin.** Nucleosomes constitute the repeating unit of a structure in nucleus called **chromatin**, thread-like stained (coloured) bodies seen in nucleus.

5 **Chromosome.** The nucleosomes in chromatin are packed to form **chromatin fibers** that are further coiled and condensed at metaphase stage of cell division to form **chromosomes**.

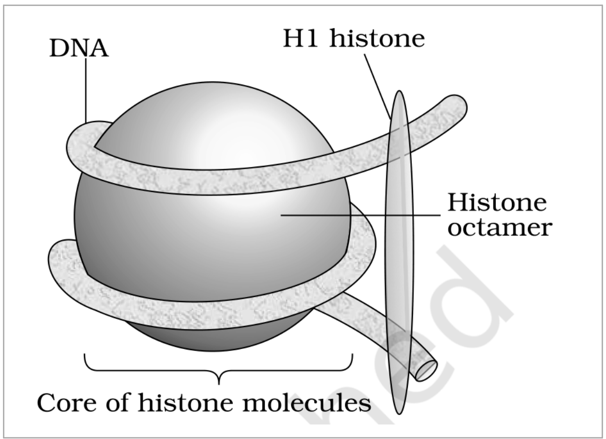


*Figure 5.4a Nucleosome. The **DNA** is wrapped around the **Histone octamer** -- the **Core of histone molecules** -- with the **H1 histone** sitting where the DNA enters and leaves; a typical nucleosome carries about 200 bp of DNA helix.*

The packaging of chromatin at higher level requires additional set of proteins that collectively are referred to as **Non-histone Chromosomal (NHC) proteins**. In a typical nucleus, some region of chromatin is loosely packed (and stains light) and is referred to as **euchromatin**. The chromatin that is more densely packed and stains dark is called as **heterochromatin**. **Euchromatin is said to be transcriptionally active chromatin, whereas heterochromatin is inactive.**

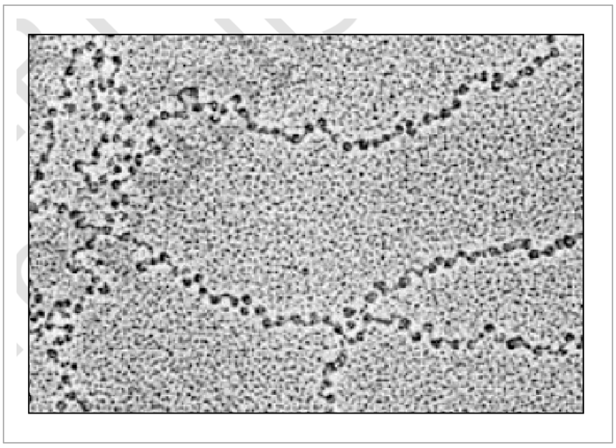


Figure 5.4b EM picture -- 'Beads-on-String'. In the electron micrograph the nucleosomes of chromatin appear as dark beads threaded on the lighter string of the intervening DNA.

ⓘ ***[NOTE] Counting the nucleosomes in a mammalian cell.*** *NCERT asks how many such beads you imagine are present in a mammalian cell. With* ***6.6×10^9 bp*** *of DNA and about* ***200 bp per nucleosome****, the estimate is 6.6×10^9 divided by 200, that is approximately* ***3.3×10^7 nucleosomes --*** *of the order of thirty million beads on the string.*

☆ ***[MEMORY AID - not in NCERT]*** *The packaging hierarchy is a single ladder of increasing order:* ***DNA to nucleosome (200 bp on a histone octamer) to chromatin to chromatin fiber to chromosome****. For staining and activity, pair them:* ***eu****chromatin is loose, light-staining and* ***active****;* ***hetero****chromatin is dense, dark-staining and* ***inactive****.*

5.2 THE SEARCH FOR GENETIC MATERIAL

Even though the discovery of **nuclein** by **Meischer** and the proposition for principles of inheritance by Mendel were almost at the same time, but the question of what molecule was actually the genetic material had not been answered. For a long time, **protein** was the favoured candidate, because proteins are chemically diverse and abundant. The experiments in this section settle the question in favour of DNA.

Where the search had reached by 1926. By **1926**, the quest to determine the mechanism for genetic inheritance had reached the **molecular level**. Previous discoveries by **Gregor Mendel**, **Walter Sutton**, **Thomas Hunt Morgan** and numerous other scientists had narrowed the search to the **chromosomes located in the nucleus** of most cells -- so the remaining question was which molecule *in* the chromosome carried the information.

5.2 Transforming Principle

In **1928**, **Frederick Griffith**, in a series of experiments with **Streptococcus pneumoniae** (bacterium responsible for pneumonia), witnessed a miraculous transformation in the bacteria. During the course of his experiment, a living organism (bacteria) had changed in physical form.

When Streptococcus pneumoniae bacteria are grown on a culture plate, some produce **smooth shiny colonies (S)** while others produce **rough colonies (R)**. This is because the **S strain bacteria have a mucous (polysaccharide) coat, while R strain does not**. Mice infected with the S strain (virulent) die from pneumonia infection but mice infected with the R strain do not develop pneumonia.

Griffith's injection	Result in mice	What it shows
S strain (live) injected into mice	Mice **die**	The S strain is virulent -- the mucous coat protects it.
R strain (live) injected into mice	Mice **live**	The R strain, lacking the coat, is non-virulent.
S strain killed by heat, injected into mice	Mice **live**	Heat-killed virulent bacteria alone cannot cause disease.
S strain killed by heat + live R strain, injected into mice	Mice **die**	The decisive result: something from the dead S cells converted live R into virulent S.

Griffith concluded that the **R strain bacteria had somehow been transformed by the heat-killed S strain bacteria**. Some **'transforming principle'**, transferred from the heat-killed S strain, had enabled the R strain to synthesise a smooth polysaccharide coat and become virulent. This must be due to the transfer of the **genetic material**. However, the **biochemical nature of genetic material was not defined** from his experiments.

5.2 Biochemical Characterisation of Transforming Principle

Prior to the work of **Oswald Avery, Colin MacLeod and Maclyn McCarty (1933-44)**, the genetic material was thought to be protein. They worked to determine the biochemical nature of the 'transforming principle' in Griffith's experiment.

They purified biochemicals (proteins, DNA, RNA, etc.) from the heat-killed S cells to see which ones could transform live R cells into S cells, and then destroyed each class of molecule in turn with a specific enzyme:

Treatment of the extract	Did transformation still occur?	Conclusion
Proteases (digest protein) and **RNases** (digest RNA) added	**Yes** -- transformation was **not** affected	Neither protein nor RNA is the transforming principle.
DNase (digests DNA) added	**No** -- DNA digestion **did** inhibit transformation	DNA is the transforming principle.

They concluded that **DNA is the hereditary material**, but not all biologists were convinced at this stage.

> ⓘ ***[NOTE] DNA versus DNase -- do not confuse the two names.*** *NCERT asks whether you can think of any difference between DNAs and DNase.* ***DNA*** *is the nucleic acid, the genetic material itself.* ***DNase*** *is an* ***enzyme*** *(a protein) that degrades DNA -- the suffix* ***-ase*** *marks an enzyme, as in protease (degrades protein) and RNase (degrades RNA). One is the substrate, the other is the scissors.*

5.2.1 The Genetic Material is DNA

The unequivocal proof that DNA is the genetic material came from the experiments of **Alfred Hershey** and **Martha Chase (1952)**. They worked with viruses that infect bacteria called **bacteriophages**.

The bacteriophage attaches to the bacteria and its genetic material then enters the bacterial cell. The bacterial cell treats the viral genetic material as if it was its own and subsequently manufactures more virus particles. Hershey and Chase worked to discover whether it was protein or DNA from the viruses that entered the bacteria. Their design turned on **radioactive labelling**:

1. **Label the two candidate molecules differently.** They grew some viruses on a medium that contained **radioactive phosphorus** and some others on medium with **radioactive sulfur**.
2. **Why those two elements.** Viruses grown in the presence of radioactive phosphorus contained **radioactive DNA but not radioactive protein**, because **DNA contains phosphorus but protein does not**. Similarly, viruses grown on radioactive sulfur contained **radioactive protein but not radioactive DNA**, because **DNA does not contain sulfur**.
3. **Infect and wait.** Radioactive phages were allowed to attach to E. coli bacteria and, as infection proceeded, the viral coats were removed from the bacteria by agitating them in a **blender**.
4. **Separate cells from coats.** The virus particles were separated from the bacteria by spinning them in a **centrifuge**.
5. **Ask which label went inside.** Bacteria which were infected with viruses that had **radioactive DNA were radioactive**, indicating that DNA was the material that passed from the virus to the bacteria. Bacteria that were infected with viruses that had **radioactive proteins were not radioactive**, indicating that proteins did not enter the bacteria from the viruses.

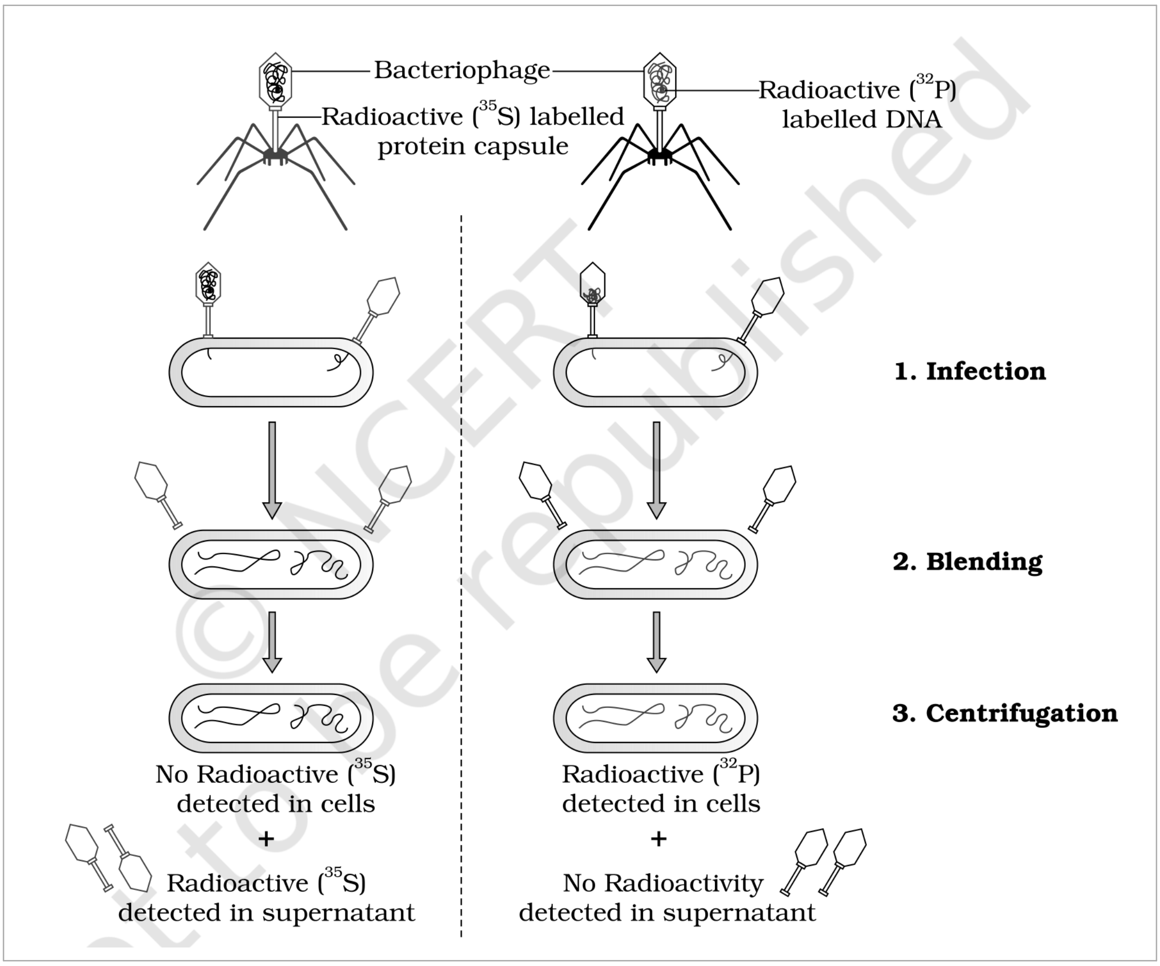


Figure 5.5 The Hershey-Chase experiment. The two panels follow one ***Bacteriophage*** *each through* ***1. Infection****,* ***2. Blending*** *and* ***3. Centrifugation****. With the* ***Radioactive (35S) labelled protein capsule****,* ***No Radioactive (35S)*** *is* ***detected in cells*** *and the* ***Radioactive (35S)*** *is* ***detected in supernatant****; with the* ***Radioactive (32P) labelled DNA****,* ***Radioactive (32P)*** *is* ***detected in cells*** *and* ***No Radioactivity*** *is* ***detected in supernatant****.*

DNA is therefore the genetic material that is passed from virus to bacteria.

☆ ***[MEMORY AID - not in NCERT]*** *The whole Hershey-Chase design rests on two element facts:* ***DNA has phosphorus but no sulfur; protein has sulfur but no phosphorus****. So* ***32P tracks DNA, 35S tracks protein --*** *and only the phosphorus label turns up inside the bacterium. Sequence of the three experiments:* ***Griffith (something transforms) - Avery, MacLeod and McCarty (that something is DNA) - Hershey and Chase (unequivocal proof)****.*

5.2.2 Properties of Genetic Material (DNA versus RNA)

From the foregoing discussion, it is clear that the debate between proteins versus DNA as the genetic material is settled in favour of DNA. It subsequently became clear, however, that in **some viruses RNA is the genetic material --** for example **Tobacco Mosaic virus** and **QB bacteriophage**. But the question of why DNA is the predominant genetic material, whereas RNA performs the dynamic functions of messenger and adapter, has to be answered from the **differences between the chemical structures of the two nucleic acid molecules**. A molecule that can act as a genetic material must fulfil the following criteria:

1. It should be able to **generate its replica** (replication).
2. It should be **chemically and structurally stable**.
3. It should provide the **scope for slow changes (mutation)** that are required for evolution.
4. It should be able to **express itself** in the form of **Mendelian characters**.

ⓘ ***[NOTE] Recall the two chemical differences between DNA and RNA.*** *NCERT asks this directly, and both answers have already appeared in 5.1.1: (i) the **sugar** -- RNA has **ribose**, with an extra -OH at the 2' position, whereas DNA has **2'-deoxyribose**; (ii) the **pyrimidine base** -- RNA has **uracil** where DNA has **thymine**. Every stability argument below follows from these two differences.*

Now judge the two nucleic acids against those criteria:

Criterion	How DNA and RNA compare
Rule-based replication	**Both** nucleic acids can direct duplication, because the **base pairing** and **complementarity** rules let each strand template the other. Proteins cannot do this, which straight away rules them out as genetic material.
Chemical stability	**DNA is more stable.** The **2'-OH group present at every nucleotide in RNA is a reactive group and makes RNA labile and easily degradable**. RNA is also known to be **catalytic, hence reactive**. Therefore **DNA is chemically more stable** and is the better store of genetic information.
Thymine versus uracil	The presence of **thymine in place of uracil** also confers **additional stability** to DNA.
Double versus single strand	The **two strands of DNA, being complementary, if separated by heating come together when appropriate conditions are provided**. Further, being **double-stranded, DNA can resist changes brought about by evolution** -- one strand can be repaired using the other as reference. NCERT stops here on purpose: the detailed discussion of this requires understanding the process of repair in DNA, which is studied in higher classes.
Mutation rate	**RNA mutates at a faster rate**. Consequently, **viruses having RNA genome and having shorter life span mutate and evolve faster**.
Ability to express	**RNA can directly code for the synthesis of proteins, hence can easily express the characters.** DNA, however, is **dependent on RNA for synthesis of proteins**.

Both DNA and RNA are therefore able to **mutate**. Putting all of this together: **DNA, being more stable, is preferred for storage of genetic information**. For the **transmission of genetic information**, **RNA is better**.

☆ ***[MEMORY AID - not in NCERT]*** *One split does the work: **DNA stores, RNA expresses**. Stability comes from what DNA lacks (no reactive 2'-OH) and what it has (thymine, a second strand for repair); RNA's reactivity is exactly what makes it a good short-lived messenger and a catalyst.*

5.3 RNA WORLD

From the foregoing discussion, an immediate question becomes evident **-- which is the first genetic material?** The properties just compared give the answer.

RNA was the first genetic material. There is now enough evidence to suggest that essential life processes (such as **metabolism**, **translation**, **splicing**, etc.) evolved around RNA. **RNA used to act as a genetic material as well as a catalyst --** there are still **some important biochemical reactions in living systems that are catalysed by RNA catalysts and not by protein enzymes**.

But **RNA being a catalyst was reactive and hence unstable**. Therefore, **DNA has evolved from RNA with chemical modifications that make it more stable**. DNA being **double stranded and having complementary strand** further **resists changes by evolving a process of repair**.

5.4 REPLICATION

While proposing the double helical structure for DNA, **Watson and Crick had immediately proposed a scheme for replication of DNA**. To quote their original statement: *"Now our model for deoxyribonucleic acid is, in effect, a pair of templates, each of which is complementary to the other. We imagine that prior to duplication the hydrogen bonds are broken, and the two chains unwind and separate. Each chain then acts as a template for the formation of a new fellow chain, so that eventually we shall have two pairs of chains, where we only had one before. Moreover, the sequence of the pairs of bases will have been duplicated exactly."*

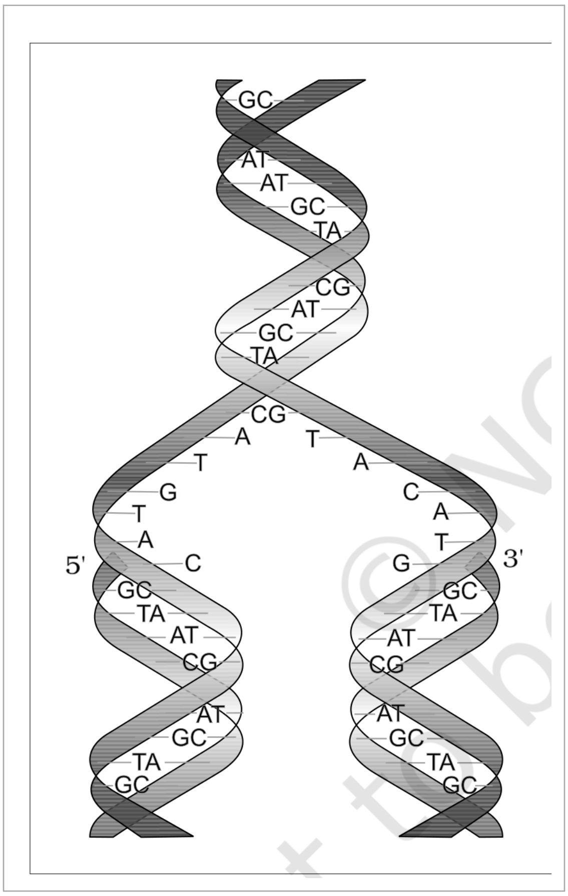


*Figure 5.6 Watson-Crick model for semiconservative DNA replication. The parental double helix unzips from the middle -- the **5'** and **3'** ends of the two separating chains are marked -- and each parental strand templates a new chain by base pairing (GC, AT, TA, CG rungs), so each daughter helix ends up with one parental and one newly synthesised strand.*

The scheme suggested that the two strands would separate and act as a template for the synthesis of new complementary strands. After the completion of replication, each DNA molecule would have **one parental and one newly synthesised strand**. This scheme was termed as **semiconservative DNA replication**.

5.4.1 The Experimental Proof

It is now proven that **DNA replicates semiconservatively**. It was shown first in **Escherichia coli** and subsequently in plants and animals. **Matthew Meselson** and **Franklin Stahl** performed the following experiment in **1958**:

1. They grew **E. coli** in a medium containing **15NH4Cl** (**15N is the heavy isotope of nitrogen**) as the only nitrogen source for **many generations**. The result was that **15N was incorporated into newly synthesised DNA (as well as other nitrogen containing compounds)**. This heavy DNA molecule could be **distinguished from normal DNA by centrifugation in a caesium chloride (CsCl) density gradient**. (Note that **15N is not a radioactive isotope**, and it can be separated from 14N only based on densities.)
2. Then they **transferred the cells into a medium with normal 14NH4Cl** and took samples at various definite time intervals as the cells multiplied, and **extracted the DNA that remained as double-stranded helices**.
3. The various samples were **separated independently on CsCl gradients** to measure the densities of DNA molecules.

> ***[NOTE] Why a density gradient separates the molecules at all.*** *NCERT asks you to recall what centrifugal force is and why a molecule with higher mass or density would sediment faster. Spinning the tube pushes material outward from the axis of rotation; the denser the molecule, the further down the CsCl gradient it travels before its density matches the surrounding solution and it stops. So **15N-15N DNA (heaviest) bands lowest, 14N-14N (lightest) bands highest, and the hybrid 15N-14N bands in between --** the position of a band is the measurement.*

The results, read generation by generation, are what make the experiment decisive:

Time in 14N medium	Density of the extracted DNA	Interpretation
After 20 minutes (one generation)	**Hybrid** or **intermediate** density	Thus, the DNA that was extracted from the culture one generation after the transfer from 15N to 14N medium had a hybrid density. Each molecule has one heavy (parental) and one light (new) strand -- exactly the semiconservative prediction.
After 40 minutes (two generations)	**Equal amounts of hybrid DNA and of light DNA**	DNA extracted after two generations was composed of equal amounts of hybrid DNA and of light DNA. The hybrid molecules each yield one hybrid and one fully light daughter.

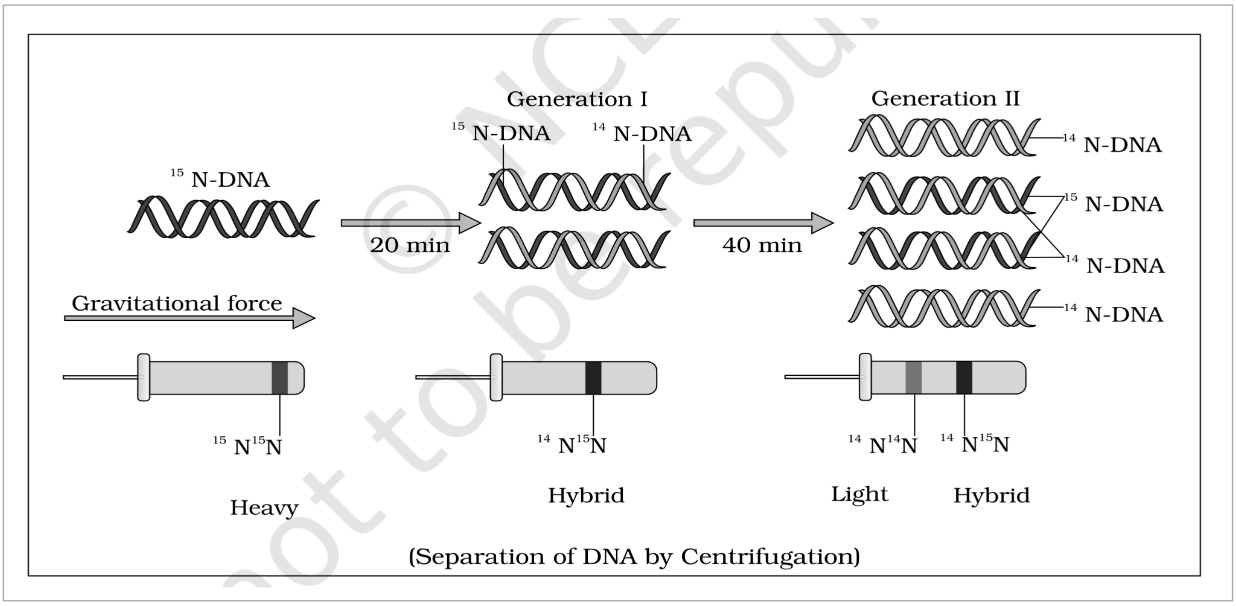


*Figure 5.7 Meselson and Stahl's Experiment. Starting from **15N-DNA**, **20 min** in 14N medium gives **Generation I** (one 15N-DNA strand with one 14N-DNA strand) and **40 min** gives **Generation II**. Under **Gravitational force** the tubes below show the bands -- 15N15N **Heavy**, 14N15N **Hybrid** and 14N14N **Light** -- the caption noting the **Separation of DNA by Centrifugation**.*

*ⓘ **[NOTE] Extending the count to 80 minutes.** NCERT asks what the proportions of light and hybrid DNA would be if E. coli were allowed to grow for 80 minutes. Since one generation is 20 minutes, 80 minutes is **four generations**, giving **16 molecules** from one. The **two original heavy strands** are conserved and remain in **2 hybrid molecules**; the other **14 are light**. So the proportion is **2 hybrid : 14 light**, that is **1/8 hybrid and 7/8 light**. The number of hybrid molecules stays fixed at two forever -- only the light ones multiply.*

Very similar experiments involving use of radioactive thymidine to detect distribution of newly synthesised DNA in the chromosomes were performed on **Vicia faba (faba beans)** by **Taylor and colleagues** in **1958**. The experiments proved that **the DNA in chromosomes also replicate semiconservatively**.

*☆ **[MEMORY AID - not in NCERT]** Meselson-Stahl in one line: **grow heavy, shift to light, watch the bands**. One generation gives **all hybrid** (this alone kills the conservative model); two generations give **half hybrid, half light** (this kills the dispersive model). The two heavy parental strands are never destroyed, so hybrid molecules stay at exactly two.*

5.4.2 The Machinery and the Enzymes

In living cells, such as E. coli, the process of replication requires a set of catalysts (**enzymes**). The main enzyme is referred to as **DNA-dependent DNA polymerase**, since it uses a DNA template to catalyse the polymerisation of deoxynucleotides. These enzymes are **highly efficient** because they have to catalyse polymerisation of a large number of nucleotides in a very short time.

> ⓘ ***[NOTE] Naming the nucleic-acid polymerases.*** *The exercises ask you to list the types of nucleic acid polymerases by the* ***chemical nature of the template (DNA or RNA)*** *and the* ***nature of the nucleic acid synthesised from it (DNA or RNA)****. The chapter's own naming convention does the work: the enzyme is named* ***[template]-dependent [product] polymerase****. That gives* ***four*** *combinations.*
>
> ***DNA-dependent DNA polymerase --*** *DNA template, DNA product: the* ***replication*** *enzyme named in this section.*
> ***DNA-dependent RNA polymerase --*** *DNA template, RNA product: the* ***transcription*** *enzyme of section 5.5.*
> ***RNA-dependent DNA polymerase --*** *RNA template, DNA product: this is the direction the chapter refers to when it notes that* ***in some viruses the flow of information is in reverse direction, that is, from RNA to DNA****.*
> ***RNA-dependent RNA polymerase --*** *RNA template, RNA product: the remaining combination, used by viruses whose genome is RNA and which must copy it as RNA.*

E. coli, that has only **4.6 x 10^6 bp**, completes the process of replication within **18 minutes**; that means the **average rate of polymerisation has to be approximately 2000 bp per second**. Not only do these enzymes have to be fast, they also have to catalyse the reaction with **high degree of energetic efficiency and accuracy**. Failure to do so would lead to mutation, and further, the **DNA molecule being very long, any mistake would result in mutations**.

The energetics and the chemistry of the reaction are worth stating explicitly:

- **Deoxyribonucleoside triphosphates serve a dual purpose**. In addition to acting as **substrates**, they provide **energy for polymerisation reaction --** the two terminal phosphates in a nucleoside triphosphate are high-energy phosphates.
- The **energetically less favourable process of separation of the two strands of the helix** means the DNA-dependent DNA polymerases **catalyse polymerisation only in one direction, that is 5' to 3'**.
- This single-direction rule creates additional complications. On one strand -- the **template with polarity 3' to 5'** -- the replication is **continuous**. On the other -- the **template with polarity 5' to 3' --** it is **discontinuous**. The **discontinuously synthesised fragments are later joined by the enzyme DNA ligase**.
- The **DNA polymerases on their own cannot initiate the process of replication**. Also, **replication does not initiate randomly at any place in DNA**. There is a definite region in E. coli DNA where the replication originates, and such a region is termed as **origin of replication**. It is due to this that **a recombinant DNA, if needs to be propagated during (particularly) in vivo, requires a vector --** a point taken up in Chapter 9.

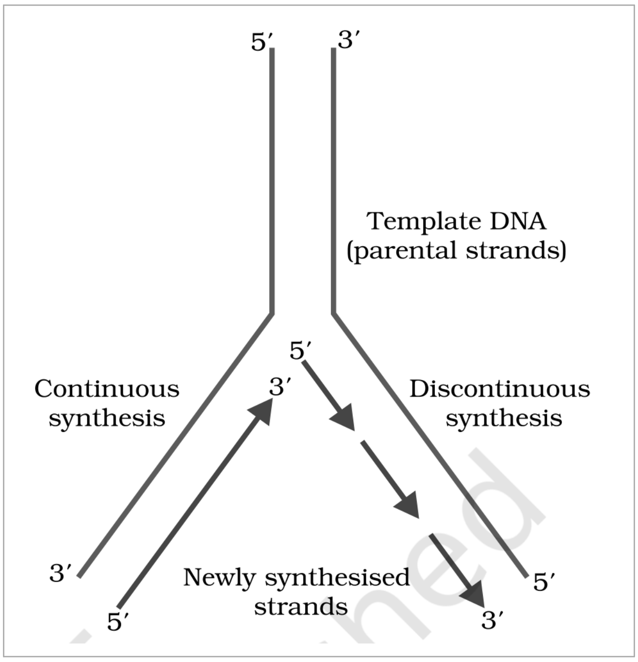


Figure 5.8 Replicating fork. The ***Template DNA (parental strands)*** *separates at the fork; on the 3' to 5' template the* ***Newly synthesised strands*** *grow by* ***Continuous synthesis****, while on the 5' to 3' template the growth is by* ***Discontinuous synthesis --*** *every arrow still running 5' to 3'.*

In eukaryotes, the replication of DNA takes place at **S-phase of the cell-cycle**. The replication of DNA and cell division cycle should be highly coordinated. A **failure in cell division after DNA replication results into**

polyploidy (a chromosomal anomaly).

☆ ***[MEMORY AID - not in NCERT]*** *Three consequences all flow from the single rule* ***polymerase works only 5' to 3'****: one new strand is* ***continuous*** *and the other* ***discontinuous****; the fragments need* ***ligase*** *to be sealed; and synthesis must start at a defined* ***origin of replication****, not anywhere. Speed benchmark for E. coli:* ***4.6 million bp in 18 minutes, about 2000 bp per second****.*

5.5 TRANSCRIPTION

- **Transcription** is the process of copying genetic information from **one strand of the DNA into RNA**. The principle of complementarity governs the process of transcription, except that here **adenosine forms base pair with uracil instead of thymine**.

However, unlike in replication, where the total DNA is duplicated, in transcription **only a segment of DNA and only one of the strands is copied into RNA**. This raises two questions that the section must answer: why both strands are not copied, and why only a segment is copied.

Why not both strands? There are two reasons. First, if both strands act as a template, they would code for RNA molecules with **different sequences** (as complementarity does not mean identical), and in turn, if they code for proteins, the sequence of amino acids in the proteins would be different. Hence, **one segment of the DNA would be coding for two different proteins**, and this would complicate the genetic information transfer machinery. Second, the **two RNA molecules if produced simultaneously would be complementary to each other**, hence would form a **double-stranded RNA**. This would prevent RNA from being translated into protein and the **exercise of transcription would become a futile one**.

5.5.1 Transcription Unit

A **transcription unit** in DNA is defined primarily by the **three regions** in the DNA:

1. A **Promoter**
2. The **Structural gene**
3. A **Terminator**

Since the two strands have **opposite polarity** and the DNA-dependent RNA polymerase also catalyse the polymerisation in only one direction, that is **5' to 3'**, the strand that has the polarity **3' to 5' acts as a template**, and is referred to as **template strand**. The other strand which has the polarity **5' to 3'** and the sequence same as RNA (except thymine at the place of uracil), is displaced during transcription. This strand (which does not code for anything) is referred to as **coding strand**. All the reference point while defining a transcription unit is made with the coding strand.

Region / strand	Polarity and role
Template strand	Polarity **3' to 5'**. This is the strand actually copied by RNA polymerase.
Coding strand	Polarity **5' to 3'**; sequence same as the RNA except thymine in place of uracil. It is **displaced** during transcription and **does not code for anything** -- despite the name. All reference points of a transcription unit are written with respect to this strand.
Promoter	Located **towards the 5' end (upstream)** of the structural gene. It is a **DNA sequence that provides binding site for RNA polymerase**, and the presence of a promoter in a transcription unit **defines the template and coding strands**. By switching its position with the terminator, the definition of coding and template strands could be reversed.
Terminator	Located **towards the 3' end (downstream)** of the coding strand, and it usually **defines the end of the process of transcription**.

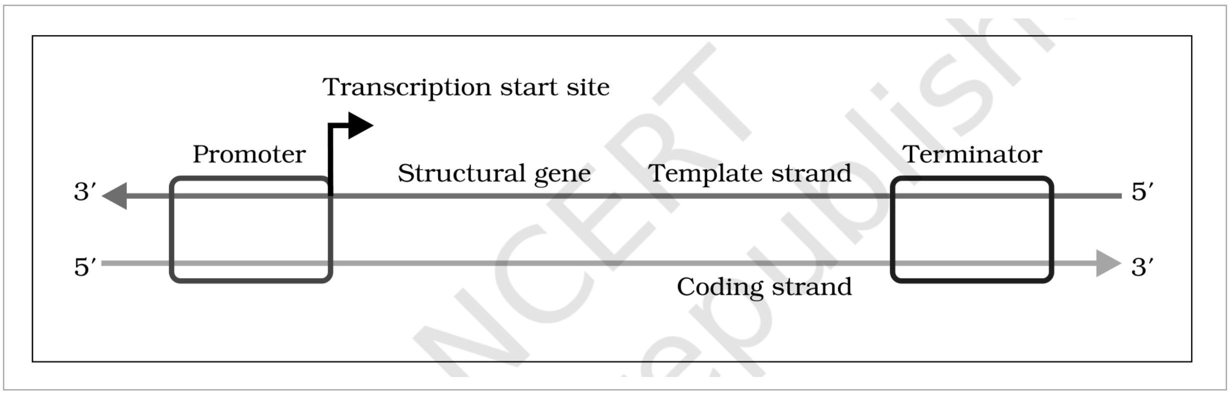


*Figure 5.9 Schematic structure of a transcription unit. The **Promoter** lies upstream with the **Transcription start site** arrow on it, then the **Structural gene**, and the **Terminator** downstream; the upper strand runs 3' to 5' and is the **Template strand**, the lower runs 5' to 3' and is the **Coding strand**.*

There are additional regulatory sequences further upstream or downstream to the promoter, and an **enhancer** is one such example.

The chapter's own worked duplex makes the two strand names concrete:

Strand	Sequence as printed
Template Strand	3'-ATGCATGCATGCATGCATGCATGC-5'
Coding Strand	5'-TACGTACGTACGTACGTACGTACG-3'

ⓘ ***[NOTE] Writing the RNA from a given DNA sequence.*** *NCERT gives a DNA duplex and asks you to write the sequence of RNA transcribed from it. Work in three moves: (i) identify the **template strand --** it is the one with 3' to 5' polarity; (ii) read it and write the **complement**; (iii) put **U wherever the rule would give T**. The quick check: the finished RNA is identical to the **coding strand** with every T replaced by U, and it runs 5' to 3'.*

ⓘ ***[NOTE] The two sequence questions in the exercises, worked.*** *Both start from the same strand, written 5' to 3': **5'-ATGCATGCATGCATGCATGCATGCATGC-3'**, that is ATGC repeated seven times.*
(a) Write the complementary strand in the 5'-to-3' direction. *Complement each base (A with T, G with C) to get 3'-TACGTACGTACGTACGTACGTACGTACG-5', then read that back the other way because the answer is wanted 5' to 3': **5'-GCATGCATGCATGCATGCATGCATGCAT-3'**. Complementing without reversing is the standard slip.*
(b) The same strand is the coding strand of a transcription unit -- write the mRNA. *The coding strand already has the RNA's sequence and polarity, so only the base substitution is needed: **5'-AUGCAUGCAUGCAUGCAUGCAUGCAUGC-3'**. No reversal here, because the mRNA matches the coding strand rather than complementing it.*

☆ ***[MEMORY AID - not in NCERT]*** *The strand names are counter-intuitive and are examined for exactly that reason: the **template strand (3' to 5') is the one that is actually copied**, while the **coding strand (5' to 3') codes for nothing** and is merely displaced. Promoter sits **upstream (5' side)**, terminator **downstream (3' side)**.*

5.5.2 Transcription Unit and the Gene

- A **gene** is defined as the **functional unit of inheritance**. This is the only place in the chapter where the gene is formally defined, so the wording is worth holding on to.

Though it is difficult to give a precise definition to gene, but since we know that the DNA sequence coding for tRNA or rRNA molecule also define a gene, we can say that **a cistron is a segment of DNA coding for a polypeptide**, and the **structural gene in a transcription unit could be said as monocistronic (mostly in eukaryotes) or polycistronic (mostly in bacteria or prokaryotes)**.

In eukaryotes, the **monocistronic structural genes have interrupted coding sequences --** the genes in eukaryotes are **split**. The **exons** are the coding sequences or expressed sequences: they are the sequences that appear in mature or processed RNA. The **exons are interrupted by introns**. The **introns** or intervening sequences **do not appear in mature or processed RNA**.

Term	Definition and where it ends up
Cistron	A **segment of DNA coding for a polypeptide**.
Monocistronic	A structural gene coding for a single polypeptide -- **mostly in eukaryotes**.
Polycistronic	A structural gene coding for more than one polypeptide -- **mostly in bacteria or prokaryotes**.
Exon	**Coding / expressed sequence**. Exons are the sequences that **appear in mature or processed RNA**.
Intron	**Intervening sequence**. Introns **do not appear in mature or processed RNA**; they interrupt the exons.

The **promoter** and the **terminator** flank the structural gene in a transcription unit. Regulatory sequences may be defined in a broad sense as **regulatory genes**, even though these sequences do not code for any RNA or protein.

> ☆ ***[MEMORY AID - not in NCERT] EX****ons are* ***EX****pressed and* ***EX****it into the mature RNA;* ***INT****rons are* ***INT****ervening and stay* ***IN****side the nucleus. For cistron counts:* ***mono****cistronic goes with* ***eukaryotes****,* ***poly****cistronic with* ***prokaryotes****.*

5.5.3 Types of RNA and the process of Transcription

In bacteria, there are **three major types of RNAs**: **mRNA (messenger RNA)**, **tRNA (transfer RNA)**, and **rRNA (ribosomal RNA)**. All three RNAs are needed to synthesise a protein in a cell, and each has a distinct job:

RNA type	Role in protein synthesis
mRNA (messenger RNA)	Provides the **template** -- the sequence to be translated.
tRNA (transfer RNA)	Brings **amino acids** and **reads the genetic code**.
rRNA (ribosomal RNA)	Plays **structural and catalytic role** during translation.

In bacteria, there is a single DNA-dependent RNA polymerase that catalyses transcription of all types of RNA. The polymerase binds to the promoter and initiates transcription (**initiation**). It uses **nucleoside triphosphates** as substrate and polymerises in a template-dependent fashion following the rule of complementarity. Somehow, it also facilitates opening of the helix and continues elongation. Only a **short stretch of RNA remains bound to the enzyme**; once the polymerases reaches the terminator region, the nascent RNA falls off, so also the RNA polymerase. This results in **termination** of transcription.

An intriguing question is how the RNA polymerase is able to catalyse all the three steps **-- initiation**, **elongation** and **termination**. The answer is that the RNA polymerase is only capable of catalysing the process of elongation. It **associates transiently with initiation-factor (sigma) and termination-factor (rho)** to initiate and terminate the transcription, respectively. **Association with these factors alters the specificity of the RNA polymerase to either initiate or terminate.**

1. **Initiation.** RNA polymerase associates with the **initiation factor (sigma)**, binds the promoter and starts transcription.
2. **Elongation.** The core RNA polymerase **--** the only step it can catalyse unaided **--** opens the helix and polymerises nucleoside triphosphates 5' to 3' following complementarity.
3. **Termination.** On reaching the terminator, and with the **termination factor (rho)**, the nascent RNA and the polymerase both fall off.

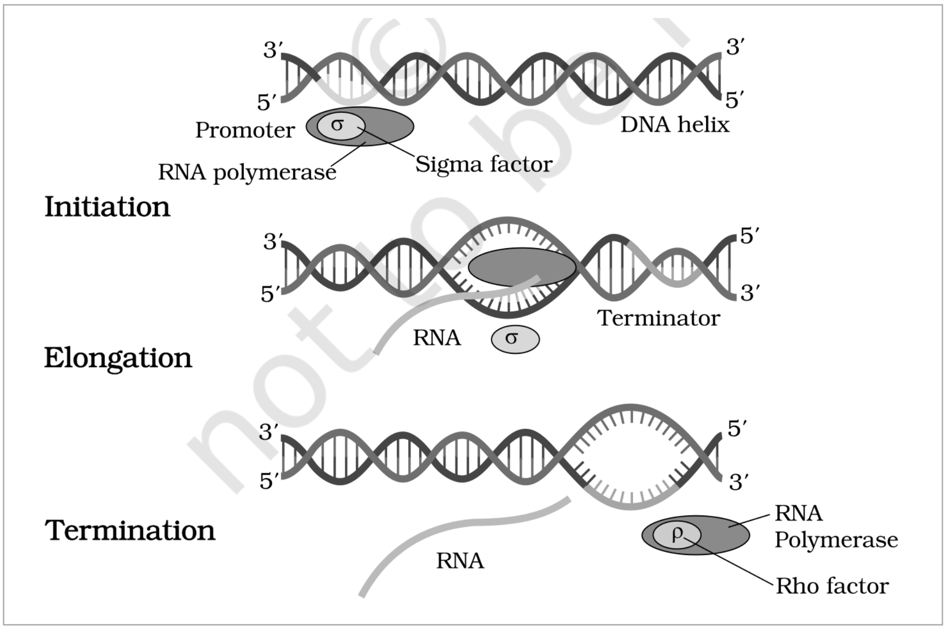


*Figure 5.10 Process of transcription in bacteria. **Initiation**: the **RNA polymerase** with the **Sigma factor** binds the **Promoter** on the **DNA helix**. **Elongation**: the helix is opened and **RNA** is polymerised as the enzyme moves towards the **Terminator**, sigma having left. **Termination**: with the **Rho factor**, both the **RNA** and the **RNA Polymerase** fall off.*

☆ *[MEMORY AID - not in NCERT] Bacterial transcription needs **one polymerase plus two factors: sigma starts** it, **rho stops** it, and the polymerase itself only elongates. For the three RNAs: **mRNA is the message, tRNA transfers the amino acid, rRNA is the ribosome --** structural and catalytic.*

In **bacteria**, since the mRNA does not require any processing to become active, and also since transcription and translation take place in the same compartment (there is no separation of cytosol and nucleus in bacteria), many times the **translation can begin much before the mRNA is fully transcribed**. Consequently, the **transcription and translation can be coupled in bacteria**.

In **eukaryotes**, there are **two additional complexities**.

(i) There are **at least three RNA polymerases in the nucleus** (in addition to the RNA polymerase found in the organelles). There is a **clear cut division of labour**:

Enzyme	What it transcribes
RNA polymerase I	Transcribes **rRNAs** -- the **28S, 18S and 5.8S** rRNAs.
RNA polymerase II	Transcribes the **precursor of mRNA**, the **heterogeneous nuclear RNA (hnRNA)**.
RNA polymerase III	Transcribes **tRNA**, **5srRNA** and **snRNAs** (small nuclear RNAs).

(ii) The second complexity is that the **primary transcripts contain both the exons and the introns and are non-functional**. Hence, it is subjected to a process called **splicing** where the **introns are removed and exons are joined in a defined order**.

hnRNA undergoes **additional processing called as capping and tailing**. In **capping**, an unusual nucleotide (**methyl guanosine triphosphate**) is added to the **5'-end** of hnRNA. In **tailing**, **adenylate residues (200-300)** are added at the **3'-end** in a **template independent manner**. It is the **fully processed hnRNA, now called mRNA**, that is **transported out of the nucleus for translation**.

1. **Transcription.** RNA polymerase II copies the split gene into **hnRNA**, which still carries every **Exon** and every **Intron**.
2. **Capping.** Methyl guanosine triphosphate is added at the **5'** end **--** the **Cap**.
3. **Tailing / Polyadenylation.** 200-300 adenylate residues are added at the **3'** end, template independently **--** the **Poly A tail**.
4. **RNA splicing.** Introns are excised and exons joined in a defined order, giving the mature **Messenger RNA**.

5 **Export.** The fully processed **3' mRNA** leaves the nucleus for translation.

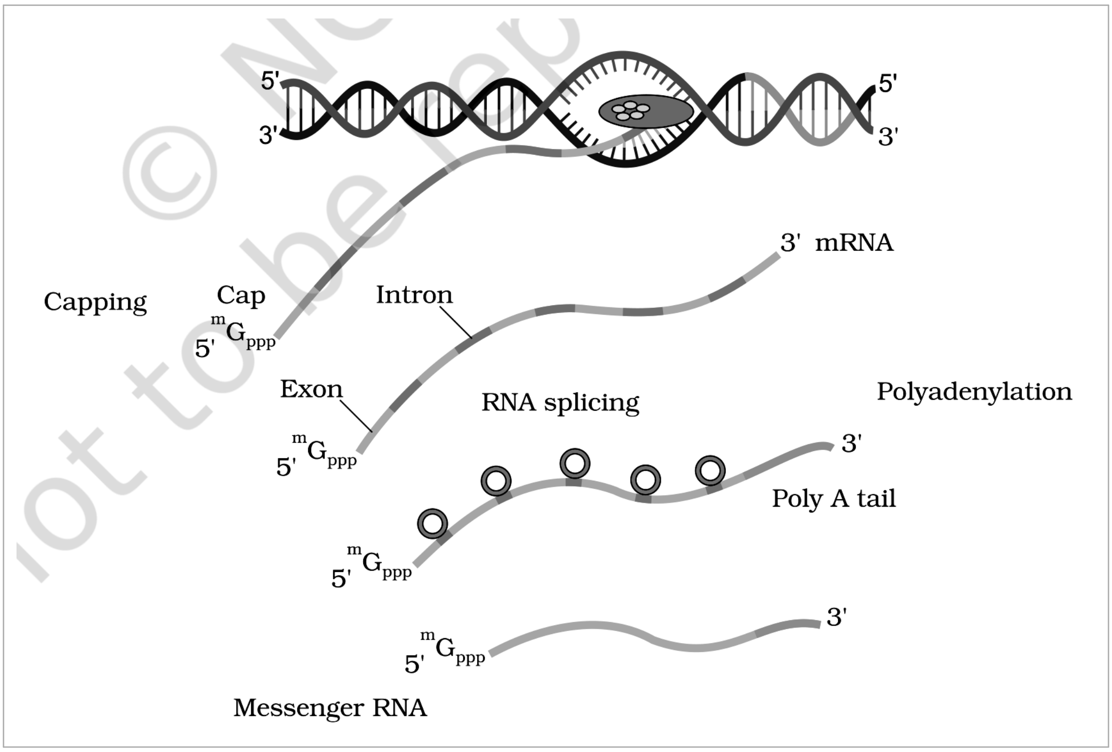


*Figure 5.11 Process of transcription in eukaryotes. The primary transcript carries **Exon** and **Intron** stretches; **Capping** adds the **Cap** at the **5'** end and **Polyadenylation** adds the **Poly A tail** at the **3'** end, then **RNA splicing** removes the introns to give the mature **Messenger RNA** -- the **3' mRNA** that is exported for translation.*

The **significance of such complexities is now beginning to be understood**. The **split-gene arrangements represent probably an ancient feature of the genome**. The **presence of introns is reminiscent of antiquity**, and the **process of splicing represents the dominance of RNA-world**. In recent times, the understanding of RNA and RNA-dependent processes in the living system have assumed more importance.

☆ ***[MEMORY AID - not in NCERT]*** *Eukaryotic processing in order:* ***cap the 5' end, tail the 3' end, then splice****. Remember the polymerase division of labour by the* ***rRNA / mRNA / tRNA*** *order of* ***I / II / III --*** *polymerase* ***II*** *is the one that matters most in this chapter because it makes* ***hnRNA****, the precursor of mRNA. And note the contrast that NEET likes: in* ***bacteria*** *transcription and translation are* ***coupled*** *(no processing, no nucleus); in* ***eukaryotes*** *they are separated by processing and by the nuclear envelope.*

5.6 GENETIC CODE

During replication and transcription a nucleic acid was copied to form another nucleic acid. Hence, these **processes are easy to conceptualise on the basis of complementarity**. The process of **translation** requires **transfer of genetic information from a polymer of nucleotides to synthesise a polymer of amino acids**. **Neither does any complementarity exist between nucleotides and amino acids, nor could any be drawn theoretically.**

There existed ample evidences, though, to support the notion that **change in nucleic acids (genetic material) were responsible for change in amino acids in proteins**. This led to the **proposition of a genetic code that could direct the sequence of amino acids during synthesis of proteins**.

If determining the biochemical nature of genetic material and the structure of DNA was very exciting, the **proposition and deciphering of genetic code were most challenging**. In a very true sense, it required involvement of **scientists from several disciplines -- physicists, organic chemists, biochemists and geneticists**.

Who	Contribution to the genetic code
George Gamow (a physicist)	Argued that since there are **only 4 bases** and if they have to **code for 20 amino acids**, the **code should constitute a combination of bases**. He suggested that in order to code for all the 20 amino acids, the **code should be made up of three nucleotides**. This was a **very bold proposition**, because a permutation combination of **4^3 (4 x 4 x 4) would generate 64 codons** -- generating **many more codons than required**.
Har Gobind Khorana	The **chemical method** he developed was instrumental in **synthesising RNA molecules with defined combinations of bases** (**homopolymers and copolymers**).
Marshall Nirenberg	His **cell-free system for protein synthesis** finally **helped the code to be deciphered**.
Severo Ochoa	His enzyme (**polynucleotide phosphorylase**) was also helpful in **polymerising RNA with defined sequences in a template independent manner** (enzymatic synthesis of RNA).

Providing proof that the codon was a triplet, was a more daunting task. Finally a **checker-board for genetic code was prepared**, which is given in **Table 5.1**.

Table 5.1 The Codons for the Various Amino Acids

1st base	2nd base: U	2nd base: C	2nd base: A	2nd base: G
U	**UUU** Phe **UUC** Phe **UUA** Leu **UUG** Leu	**UCU** Ser **UCC** Ser **UCA** Ser **UCG** Ser	**UAU** Tyr **UAC** Tyr **UAA (Stop)** **UAG (Stop)**	**UGU** Cys **UGC** Cys **UGA (Stop)** **UGG** Trp
C	**CUU** Leu **CUC** Leu **CUA** Leu **CUG** Leu	**CCU** Pro **CCC** Pro **CCA** Pro **CCG** Pro	**CAU** His **CAC** His **CAA** Gln **CAG** Gln	**CGU** Arg **CGC** Arg **CGA** Arg **CGG** Arg
A	**AUU** Ile **AUC** Ile **AUA** Ile **AUG** Met	**ACU** Thr **ACC** Thr **ACA** Thr **ACG** Thr	**AAU** Asn **AAC** Asn **AAA** Lys **AAG** Lys	**AGU** Ser **AGC** Ser **AGA** Arg **AGG** Arg
G	**GUU** Val **GUC** Val **GUA** Val **GUG** Val	**GCU** Ala **GCC** Ala **GCA** Ala **GCG** Ala	**GAU** Asp **GAC** Asp **GAA** Glu **GAG** Glu	**GGU** Gly **GGC** Gly **GGA** Gly **GGG** Gly

> ⓘ ***[NOTE] How to read the checker-board.*** *Take the codon's **first base from the row**, its **second base from the column**, and find the line inside that cell whose **third base** matches **--** the third base always runs **U, C, A, G** down the cell. So **AUG** is row **A**, column **U**, third line **G: Met**. Reading the whole board confirms the counts quoted in the salient features below **-- 64 codons** in all, of which **61 code for amino acids** and **3 (UAA, UAG, UGA) are stop codons**.*

The **salient features of genetic code** are as follows:

Feature	Statement
(i) Triplet	The **codon is triplet**. **61 codons code for amino acids** and **3 codons do not code for any amino acids**, hence they **function as stop codons**.
(ii) Degenerate	**Some amino acids are coded by more than one codon**, hence the **code is degenerate**.
(iii) Contiguous	The **codon is read in mRNA in a contiguous fashion**. **There are no punctuations.**
(iv) Nearly universal	The **code is nearly universal**: for example, **from bacteria to human UUU would code for Phenylalanine (phe)**. **Some exceptions to this rule have been found in mitochondrial codons, and in some protozoans.**
(v) AUG is dual	**AUG has dual functions.** It **codes for Methionine (met)**, and it also **act as initiator codon**.
(vi) Stop codons	**UAA, UAG, UGA are stop terminator codons.**

> *[NOTE] **Worked exercise (NCERT, in-body).** If following is the sequence of nucleotides in mRNA, predict the sequence of amino acid coded by it (take help of the checkerboard): **-AUG UUU UUC UUC UUU UUU UUC-**. Reading Table 5.1 codon by codon: AUG = **Met**, UUU = **Phe**, UUC = **Phe**, UUC = **Phe**, UUU = **Phe**, UUU = **Phe**, UUC = **Phe**. So the peptide is **Met-Phe-Phe-Phe-Phe-Phe-Phe**.*
>
> *Now try the opposite. Following is the sequence of amino acids coded by an mRNA. Predict the nucleotide sequence in the RNA: **Met-Phe-Phe-Phe-Phe-Phe-Phe**. Do you face any difficulty in predicting the opposite? Yes **--** and that is the point of the question. **Met** is unambiguous (only **AUG**), but each **Phe** could be **UUU or UUC**, so six positions have two choices each and the answer is not unique.*
>
> *Can you now correlate which two properties of genetic code you have learnt? The forward direction works unambiguously because the code is a **triplet** read **contiguously without punctuation**; the reverse direction is ambiguous because the code is **degenerate**. Degeneracy is why translation is one-way readable: **codon to amino acid is certain, amino acid to codon is not**.*

> *[MEMORY AID - not in NCERT]** Fix the numbers: **4 bases, triplet code, 4^3 = 64 codons, 61 coding + 3 stop, for 20 amino acids**. The three stops are **UAA, UAG, UGA --** remember them as **U-A-A / U-A-G / U-G-A**, all beginning with U. **AUG** is the one codon with **two jobs**: **start** and **Met**. And keep the two contrasted properties straight: **degenerate** = one amino acid, many codons; **unambiguous** = one codon, never two amino acids.*

5.6.1 Mutations and Genetic Code

The **relationships between genes and DNA are best understood by mutation studies**. You have studied about **mutation and its effect in Chapter 4**.

Effects of **large deletions and rearrangements** in a segment of DNA are easy to comprehend. It may result in **loss or gain of a gene and so a function**. The effect of **point mutations** will be explained here.

- A **classical example of point mutation** is a **change of single base pair in the gene for beta globin chain** that results in the **change of amino acid residue glutamate to valine**. It results into a diseased condition called as **sickle cell anemia**.

Effect of point mutations that **inserts or deletes a base** in a structural gene can be better understood by the following simple example. Consider a statement that is made up of the following words **each having three letters like genetic code**:

What is done to the sentence	How it now reads
The original statement (every word a triplet)	**RAM HAS RED CAP**
Insert one letter B in between HAS and RED, and rearrange	**RAM HAS BRE DCA P**
Insert two letters at the same place, say **BI**	**RAM HAS BIR EDC AP**
Insert three letters together, say **BIG**	**RAM HAS BIG RED CAP**
Delete one letter (R), and rearrange	**RAM HAS EDC AP**
Delete two letters (R, E), and rearrange	**RAM HAS DCA P**
Delete three letters (R, E, D), and rearrange	**RAM HAS CAP**

The **conclusion from the above exercise is very obvious**. **Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion.** However, such mutations are referred to as **frameshift insertion or deletion mutations**.

Insertion or deletion of three or its multiple bases insert or delete in one or multiple codon hence **one or multiple amino acids**, and the **reading frame remains unaltered from that point onwards**.

> *[MEMORY AID - not in NCERT]** The whole point of **RAM HAS RED CAP** is the **multiple-of-three rule**: add or remove **1 or 2** bases and every codon downstream is garbled **--** a **frameshift**; add or remove **3 or a multiple of 3** and you only gain or lose whole amino acids, with the **reading frame intact**. Note the contrast NEET tests: **sickle cell anemia is a substitution** (one base pair changed, glutamate to valine), **not** a frameshift.*

5.6.2 tRNA - the Adapter Molecule

From the very beginning of the proposition of code, it was **clear to Francis Crick** that there has to be a **mechanism to read the code and also to link it to the amino acids**, because **amino acids have no structural specialities to read the code uniquely**. He postulated the **presence of an adapter molecule** that would **on one hand read the code and on other hand would bind to specific amino acids**.

The **tRNA, then called sRNA (soluble RNA)**, was **known before the genetic code was postulated**. However, its **role as an adapter molecule was assigned much later**.

- **tRNA has an anticodon loop that has bases complementary to the code**, and it also has an **amino acid acceptor end to which it binds to amino acids**. **tRNAs are specific for each amino acid.**

For **initiation**, there is **another specific tRNA** that is referred to as **initiator tRNA**. **There are no tRNAs for stop codons.**

In Figure 5.12, the **secondary structure of tRNA** has been depicted that **looks like a clover-leaf**. In **actual structure, the tRNA is a compact molecule which looks like inverted L**.

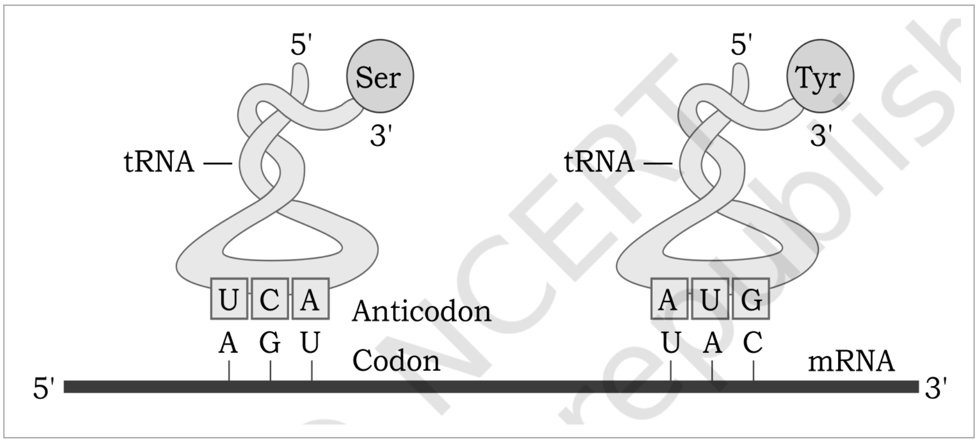


*Figure 5.12 tRNA -- the adapter molecule. The clover-leaf secondary structure of **tRNA** carries its amino acid at the acceptor end (shown for **Ser** and **Tyr**) and its **Anticodon** on the **anticodon loop**, which pairs with the matching **Codon** on the **mRNA** running **5'** to **3'**.*

☆ ***[MEMORY AID - not in NCERT]** tRNA is the **adapter**: **two ends, two jobs --** the **anticodon loop reads the mRNA codon** by complementarity, the **amino acid acceptor end carries the amino acid**. Two shape facts are examined against each other: the **secondary** structure is a **clover-leaf**, the **actual** three-dimensional structure is an **inverted L**. And remember the two absences: **no tRNA for stop codons**, but there **is** a special **initiator tRNA**.*

5.7 TRANSLATION

- **Translation** refers to the **process of polymerisation of amino acids to form a polypeptide**. The **order and sequence of amino acids are defined by the sequence of bases in the mRNA**.

The **amino acids are joined by a bond which is known as a peptide bond**. **Formation of a peptide bond requires energy.** Therefore, in the **first phase itself amino acids are activated in the presence of ATP and linked to their cognate tRNA --** a process commonly called as **charging of tRNA** or **aminoacylation of tRNA** to be more specific.

If **two such charged tRNAs are brought close enough**, the **formation of peptide bond between them would be favoured energetically**. The **presence of a catalyst would enhance the rate of peptide bond formation**.

- The **cellular factory responsible for synthesising proteins is the ribosome**. The ribosome consists of **structural RNAs and about 80 different proteins**.

In its **inactive state, it exists as two subunits: a large subunit and a small subunit**. When the **small subunit encounters an mRNA**, the **process of translation of the mRNA to protein begins**. There are **two sites in the large subunit**, for **subsequent amino acids to bind to** and thus, be **close enough to each other for the formation of a peptide bond**. The **ribosome also acts as a catalyst (23S rRNA in bacteria is the enzyme -- ribozyme) for the formation of peptide bond**.

Region of the mRNA	What it is
Translational unit	A **translational unit in mRNA is the sequence of RNA that is flanked by the start codon (AUG) and the stop codon and codes for a polypeptide**.
Untranslated regions (UTR)	An mRNA also has some **additional sequences that are not translated** and are referred as **untranslated regions (UTR)**. The **UTRs are present at both 5'-end (before start codon) and at 3'-end (after stop codon)**. They are **required for efficient translation process**.

1. **Charging / aminoacylation.** Amino acids are **activated in the presence of ATP** and **linked to their cognate tRNA**.
2. **Initiation.** The **ribosome binds to the mRNA at the start codon (AUG)** that is **recognised only by the initiator tRNA**.
3. **Elongation.** Complexes of an **amino acid linked to tRNA** sequentially bind the **appropriate codon in mRNA by forming complementary base pairs with the tRNA anticodon**; the **ribosome moves from codon to codon along the mRNA** and **amino acids are added one by one**, translated into **Polypeptide** sequences dictated by DNA and represented by mRNA.
4. **Termination.** A **release factor binds to the stop codon**, **terminating translation** and **releasing the complete polypeptide from the ribosome**.

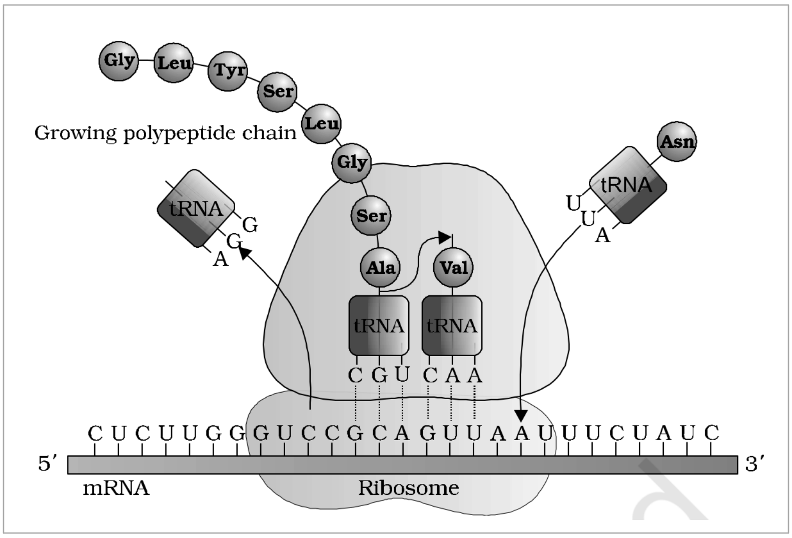


*Figure 5.13 Translation. The **Ribosome** moves along the **mRNA** from **5'** to **3'** while each **tRNA** delivers its amino acid to the **Growing polypeptide chain** -- here the residues **Gly**, **Leu**, **Tyr**, **Ser**, **Ala**, **Val** and **Asn**.*

> *ⓘ **[NOTE] Why translation is evidence for the RNA world.** Look at what does the work in this section: the **message is RNA** (mRNA), the **adapter is RNA** (tRNA), and the **catalyst is RNA** (the 23S rRNA ribozyme). Proteins are the product, not the machinery. **Translation is a process that has evolved around RNA, indicating that life began around RNA.***

> *☆ **[MEMORY AID - not in NCERT]** Two ribosome facts answer the common two-mark question "list two essential roles of the ribosome": it provides the **two binding sites in the large subunit** that hold successive amino acids close enough to react, and its **23S rRNA acts as a ribozyme** catalysing the peptide bond. Remember **charging comes before initiation --** ATP is spent activating the amino acid, not making the peptide bond directly. UTRs sit **outside** the start and stop codons and are **not translated**, yet are needed for efficient translation.*

5.8 REGULATION OF GENE EXPRESSION

- **Regulation of gene expression** refers to a **very broad term that may occur at various levels**. Considering that **gene expression results in the formation of a polypeptide**, it can be **regulated at several levels**.

In **eukaryotes**, the regulation could be exerted at:

Level	What is regulated there
(i) Transcriptional level	Formation of the **primary transcript**.

Level	What is regulated there
(ii) Processing level	**Regulation of splicing.**
(iii) Transport	**Transport of mRNA from nucleus to the cytoplasm.**
(iv) Translational level	**Translation** of the mRNA itself.

The **genes in a cell are expressed to perform a particular function or a set of functions**. For example, if an enzyme called **beta-galactosidase** is synthesised by **E. coli**, it is used to **catalyse the hydrolysis of a disaccharide, lactose into galactose and glucose**; the **bacteria use them as a source of energy**. Hence, if the **bacteria do not have lactose around them** to be utilised for energy source, they would **no longer require the synthesis of the enzyme beta-galactosidase**.

- Therefore, in simple terms, it is the **metabolic, physiological or environmental conditions that regulate the expression of genes**. The **development and differentiation of embryo into adult organisms** are also a result of the **coordinated regulation of expression of several sets of genes**.

> ⓘ ***[NOTE] Why regulation is not optional. Since transcription and translation are energetically very expensive processes, these have to be tightly regulated.*** *A cell that transcribed and translated every gene all the time would spend its energy budget producing enzymes for substrates that are not there.*

In **prokaryotes, control of the rate of transcriptional initiation is the predominant site for control of gene expression**. In a transcription unit, the **activity of RNA polymerase at a given promoter is in turn regulated by interaction with accessory proteins**, which **affect its ability to recognise start sites**. These **regulatory proteins can act both positively (activators) and negatively (repressors)**.

- The **accessibility of promoter regions of prokaryotic DNA** is in many cases **regulated by the interaction of proteins with sequences termed operators**. The **operator region is adjacent to the promoter elements in most operons** and in most cases the **sequences of the operator bind a repressor protein**.

Each operon has its specific operator and specific repressor. For example, **lac operator is present only in the lac operon** and it **interacts specifically with lac repressor only**.

> ☆ ***[MEMORY AID - not in NCERT]*** *Note the **qualifier**, because it is the difference between a right and a wrong answer: it is **in prokaryotes** that **transcriptional initiation** is the **predominant** control point. In eukaryotes there are **four** levels **-- transcription, processing, transport, translation**. Regulatory proteins come in two signs: **activators (positive)** and **repressors (negative)**.*

5.8.1 The Lac operon

The **elucidation of the lac operon** was also a result of a **close association between a geneticist, Francois Jacob and a biochemist, Jacque Monod**. They were the **first to elucidate a transcriptionally regulated system**.

- In **lac operon** (here **lac refers to lactose**), a **polycistronic structural gene is regulated by a common promoter and regulatory genes**. Such arrangement is **very common in bacteria** and is referred to as **operon**. **Lac operon is the prototype operon in bacteria**, which codes for genes **responsible for metabolism of lactose**.

To name few such examples: **lac operon, trp operon, ara operon, his operon, val operon**, etc.

The **lac operon consists of one regulatory gene (the i gene --** here the **term i does not refer to inducer, rather it is derived from the word inhibitor**) **and three structural genes (z, y, and a)**.

Gene	Product and its job
i gene (regulatory)	Codes for the **repressor of the lac operon**. It is synthesised **all-the-time (constitutively)**.
z gene	Codes for **beta-galactosidase (beta-gal)**, which is **primarily responsible for the hydrolysis of the disaccharide, lactose into its monomeric units, galactose and glucose**.
y gene	Codes for **permease**, which **increases permeability of the cell to beta-galactosides**.
a gene	Encodes a **transacetylase**.

Hence, **all the three gene products in lac operon are required for metabolism of lactose**. In **most other operons as well, the genes present in the operon are needed together to function in the same or related**

metabolic pathway.

- **Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the operon.** Hence, it is termed as **inducer**.

In the **absence of a preferred carbon source such as glucose**, if **lactose is provided in the growth medium** of the bacteria, the **lactose is transported into the cells through the action of permease**. (Remember, a **very low level of expression of lac operon has to be present in the cell all the time**, otherwise **lactose cannot enter the cells**.) The lactose then **induces the operon** in the following manner.

1 The **repressor of the operon is synthesised (all-the-time -- constitutively) from the i gene**, as **Repressor mRNA** and then **Repressor** protein.

2 **In absence of inducer:** the **Repressor binds to the operator region (o) and prevents RNA polymerase from transcribing the operon**.

3 **In presence of inducer:** in the presence of an **inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer --** an **Inactive repressor**.

4 This **allows RNA polymerase access to the promoter and transcription proceeds**, giving **lac mRNA**.

5 **Transcription** then **Translation** of that polycistronic message yields **beta-galactosidase**, **permease** and **transacetylase**.

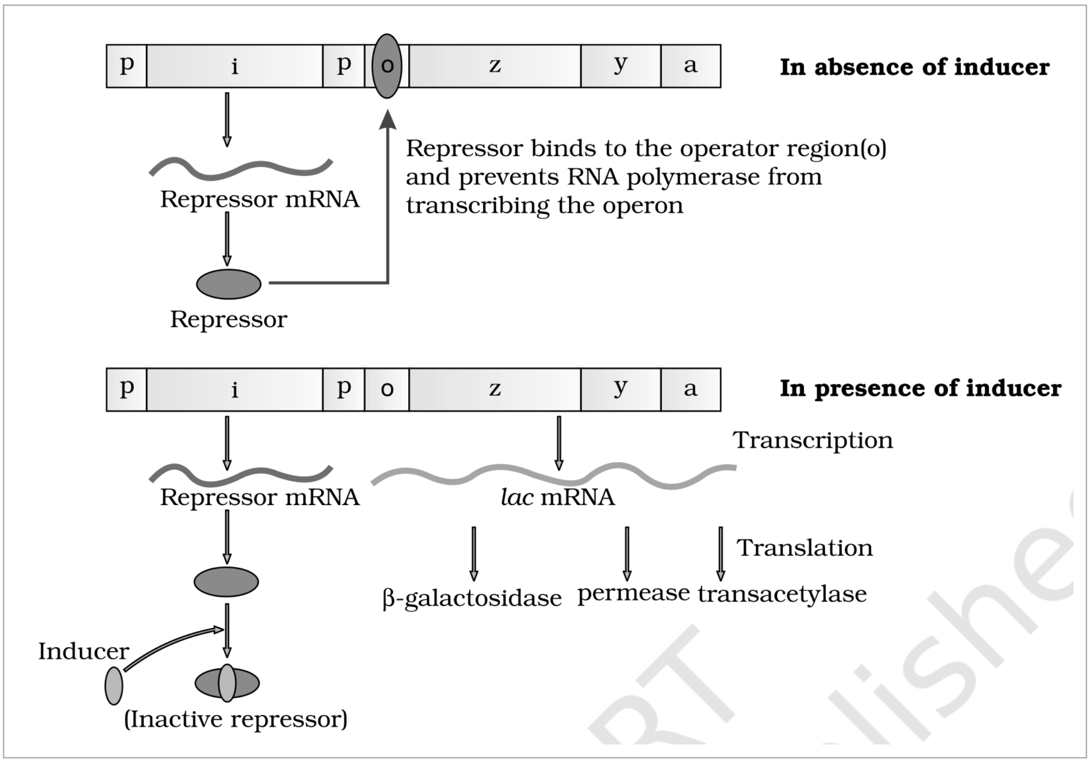


*Figure 5.14 The lac operon. **In absence of inducer**, the **Repressor mRNA** from the i gene gives the **Repressor**, and the **Repressor binds to the operator region (o) and prevents RNA polymerase from transcribing the operon**. **In presence of inducer**, the **Inducer** gives an **Inactive repressor**, so **lac mRNA** is made and **Transcription** followed by **Translation** yields **beta-galactosidase**, **permease** and **transacetylase**.*

Essentially, **regulation of lac operon can also be visualised as regulation of enzyme synthesis by its substrate**. Remember, **glucose or galactose cannot act as inducers for lac operon**. **Regulation of lac operon by repressor is referred to as negative regulation. Lac operon is under control of positive regulation as well, but it is beyond the scope of discussion at this level.**

> ⓘ ***[NOTE] Why the operon switches itself off again -- and the answer to "can you think for how long the lac operon would be expressed in the presence of lactose?"** Only **as long as the lactose lasts**. Put the facts of this section in a line: **lactose is the inducer**, and **lactose is also the substrate of beta-galactosidase**, the very enzyme the operon switches on. So the induced enzyme **consumes the inducer --** hydrolysing it to **galactose and glucose**, neither of which **can act as an inducer**. As the lactose is used up there is nothing left to **inactivate the repressor**, the repressor **binds the operator again**, and **transcription stops**. The operon is therefore self-limiting: it stays on only while its own substrate is present.*

> ☆ ***[MEMORY AID - not in NCERT]** Gene letters, in order: **i-z-y-a** = **inhibitor (repressor), beta-gal, permease, transacetylase**. The **i stands for inhibitor, not inducer --** a favourite trap. The **inducer is lactose (or allolactose)**; **glucose and galactose are not inducers**. Repressor **on** the operator = operon **off**; inducer **inactivates** the repressor = operon **on**. This repressor-based control is **negative regulation**.*

5.9 HUMAN GENOME PROJECT

In the preceding sections you have learnt that it is the **sequence of bases in DNA that determines the genetic information of a given organism**. In other words, **genetic make-up of an organism or an individual lies in the DNA sequences**. If **two individuals differ, then their DNA sequences should also be different, at least at some places**. These assumptions led to the **quest of finding out the complete DNA sequence of human genome**.

With the establishment of **genetic engineering techniques** where it was possible to **isolate and clone any piece of DNA** and availability of **simple and fast techniques for determining DNA sequences**, a very **ambitious project of sequencing human genome was launched in the year 1990**.

- **Human Genome Project (HGP) was called a mega project.** You can imagine the magnitude and the requirements for the project if we simply define the aims of the project as follows.

What makes it a mega project	The figure NCERT gives
Size of the genome	Human genome is said to have approximately **3 x 10^9 bp**.
Cost per base	**US $ 3 per bp** (the estimated cost in the beginning).
Total estimated cost	Approximately **9 billion US dollars**.
Storage, if printed	If each page of a book contained **1000 letters** and each book contained **1000 pages**, then **3300 such books** would be required to store the information of DNA sequence **from a single human cell**.
Computation	The **enormous amount of data** expected to be generated also necessitated the use of **high speed computational devices for data storage and retrieval, and analysis**.
Duration and coordination	A **13-year project coordinated by the U.S. Department of Energy and the National Institute of Health**. During the early years, the **Wellcome Trust (U.K.)** became a major partner; additional contributions came from **Japan, France, Germany, China** and others. The project was **completed in 2003**.

- **HGP was closely associated with the rapid development of a new area in biology called Bioinformatics.**

> ⓘ ***[NOTE] What Bioinformatics does.** NCERT names the field but does not say what it is, and the exercises ask you to describe it **--** so read it off the need that created it. The project generated an **enormous amount of data** requiring **high speed computational devices for data storage and retrieval, and analysis**, and two of the HGP's own stated goals were to **store this information in databases** and to **improve tools for data analysis**. **Bioinformatics is that area of biology: the use of computational methods and databases to store, retrieve and analyse biological sequence information.***

Goals: Goals of HGP

Some of the **important goals of HGP** were as follows:

#	Goal
(i)	**Identify all the approximately 20,000-25,000 genes in human DNA;**
(ii)	**Determine the sequences of the 3 billion chemical base pairs that make up human DNA;**

#	Goal
(iii)	**Store this information in databases;**
(iv)	**Improve tools for data analysis;**
(v)	**Transfer related technologies to other sectors, such as industries;**
(vi)	**Address the ethical, legal, and social issues (ELSI) that may arise from the project.**

Knowledge about the effects of DNA variations among individuals can lead to revolutionary new ways to diagnose, treat and someday prevent the thousands of disorders that affect human beings. Besides providing clues to understanding human biology, learning about **non-human organisms DNA sequences** can lead to an understanding of their **natural capabilities** that can be applied toward solving challenges in **health care, agriculture, energy production, environmental remediation**. **Many non-human model organisms**, such as **bacteria, yeast, Caenorhabditis elegans (a free living non-pathogenic nematode), Drosophila (the fruit fly), plants (rice and Arabidopsis)**, etc., have **also been sequenced**.

Methodologies : The **methods involved two major approaches**.

Approach	What it sequenced
Expressed Sequence Tags (ESTs)	One approach focused on **identifying all the genes that are expressed as RNA**.
Sequence Annotation	The other took the **blind approach of simply sequencing the whole set of genome that contained all the coding and non-coding sequence**, and **later assigning different regions in the sequence with functions**.

1. **Isolate and fragment.** The **total DNA from a cell is isolated** and **converted into random fragments of relatively smaller sizes** (recall DNA is a very long polymer, and there are **technical limitations in sequencing very long pieces of DNA**).
2. **Clone.** The fragments are **cloned in suitable host using specialised vectors**. The **cloning resulted into amplification of each piece of DNA fragment** so that it subsequently **could be sequenced with ease**. The **commonly used hosts were bacteria and yeast**, and the vectors were called **BAC (bacterial artificial chromosomes)** and **YAC (yeast artificial chromosomes)**.
3. **Sequence.** The fragments were sequenced using **automated DNA sequencers** that worked on the **principle of a method developed by Frederick Sanger**. (Remember, **Sanger is also credited for developing method for determination of amino acid sequences in proteins**.)
4. **Assemble.** These sequences were then **arranged based on some overlapping regions present in them**. This required **generation of overlapping fragments for sequencing**. **Alignment of these sequences was humanly not possible**, therefore **specialised computer based programs were developed**.
5. **Annotate and assign.** These sequences were **subsequently annotated and were assigned to each chromosome**. The **sequence of chromosome 1 was completed only in May 2006 --** this was the **last of the 24 human chromosomes (22 autosomes and X and Y) to be sequenced**.

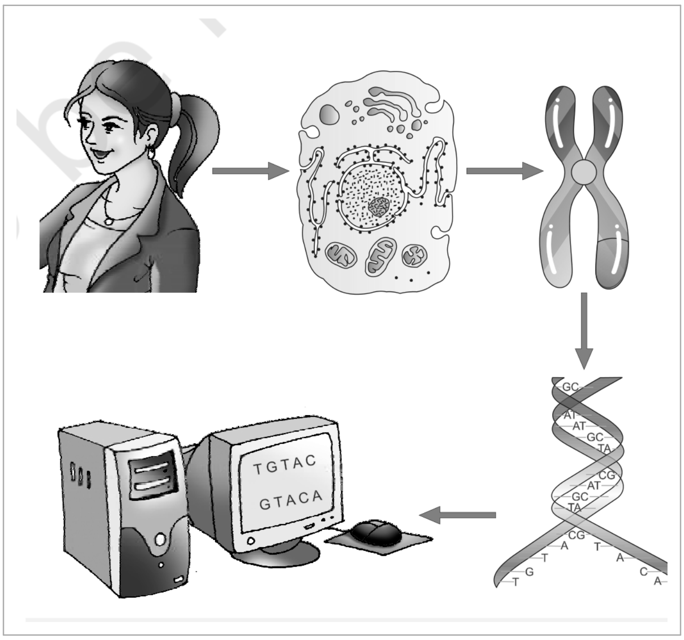


Figure 5.15 Human Genome Project -- the sequencing workflow, from total DNA through random fragments cloned in BAC and YAC vectors, to automated sequencing and computer-based alignment of overlapping fragments.

Another **challenging task was assigning the genetic and physical maps on the genome**. This was generated using information on **polymorphism of restriction endonuclease recognition sites**, and some **repetitive DNA sequences known as microsatellites**.

> ☆ ***[MEMORY AID - not in NCERT]*** *The mega-project numbers worth carrying:* ***launched 1990, completed 2003, a 13-year project****;* ***3 x 10^9 bp*** *at* ***US $3 per bp*** *= about* ***9 billion dollars****;* ***3300 books*** *to print one cell's sequence.* ***Chromosome 1 was last****, finished* ***May 2006 --*** *after the project itself was declared complete. Two methodologies, two names:* ***ESTs*** *= genes expressed as RNA only;* ***Sequence Annotation*** *= sequence everything, assign function later.* ***Sanger*** *for the sequencing method,* ***BAC and YAC*** *for the vectors.*

5.9.1 Salient Features of Human Genome

Some of the **salient observations drawn from human genome project** are as follows:

#	Observation
(i)	The **human genome contains 3164.7 million bp**.
(ii)	The **average gene consists of 3000 bases**, but **sizes vary greatly**, with the **largest known human gene being dystrophin at 2.4 million bases**.
(iii)	The **total number of genes is estimated at 30,000** -- much **lower than previous estimates of 80,000 to 1,40,000 genes**. **Almost all (99.9 per cent) nucleotide bases are exactly the same in all humans.**
(iv)	The **functions are unknown for over 50 per cent of the discovered genes**.
(v)	**Less than 2 per cent of the genome codes for proteins.**
(vi)	**Repeated sequences make up very large portion of the human genome.**
(vii)	**Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times.** They are **thought to have no direct coding functions**, but they **shed light on chromosome structure, dynamics and evolution**.
(viii)	**Chromosome 1 has most genes (2968)**, and the **Y has the fewest (231)**.
(ix)	Scientists have identified about **1.4 million locations where single-base DNA differences (SNPs -- single nucleotide polymorphism, pronounced as 'snips') occur in humans**. This information promises to **revolutionise the processes of finding chromosomal locations for disease-associated sequences and tracing human history**.

☆ ***[MEMORY AID - not in NCERT]** The numbers most often asked: **3164.7 million bp; average gene 3000 bases; dystrophin is the largest at 2.4 million bases; about 30,000 genes; 99.9 per cent of bases identical between humans; functions unknown for over 50 per cent** of genes; **less than 2 per cent codes for protein; chromosome 1 has the most genes (2968), Y the fewest (231); 1.4 million SNPs**. Do not confuse **chromosome 1 has the most genes** with **chromosome 1 was sequenced last --** both are true and both are examined.*

5.9.2 Applications and Future Challenges

Deriving meaningful knowledge from the DNA sequences will define research through the coming decades leading to our understanding of biological systems. This **enormous task will require the expertise and creativity of tens of thousands of scientists from varied disciplines in both the public and private sectors worldwide**.

One of the **greatest impacts of having the HG sequence** may well be **enabling a radically new approach to biological research**. **In the past, researchers studied one or a few genes at a time.** With **whole-genome sequences and new high-throughput technologies**, we can **approach questions systematically and on a much broader scale**. They can **study all the genes in a genome**, for example, **all the transcripts in a particular tissue or organ or tumor**, or **how tens of thousands of genes and proteins work together in interconnected networks to orchestrate the chemistry of life**.

☆ ***[MEMORY AID - not in NCERT]** The shift the HGP caused is the examinable idea here: **from one-or-a-few genes at a time to whole genomes at once**, made possible by **high-throughput technologies**. That is also the answer to why **Bioinformatics** had to be invented alongside it.*

5.10 DNA FINGERPRINTING

- As stated in the preceding section, **99.9 per cent of base sequence among humans is the same**. Assuming human genome as **3 x 10^9 bp**, in how many base sequences would there be differences? It is **these differences in sequence of DNA which make every individual unique in their phenotypic appearance**.

If one aims to **find out genetic differences between two individuals or among individuals of a population**, **sequencing the DNA every time would be a daunting and expensive task**. **Imagine trying to compare two sets of 3 x 10^6 base pairs.**

ⓘ ***[NOTE] A printing inconsistency in the source, so you do not think you misread.** This section quotes the human genome as **3 x 10^9 bp** and then, two sentences later, as **3 x 10^6 base pairs**. Both are reproduced above exactly as NCERT prints them. The figure established by the Human Genome Project in the preceding section is **3 x 10^9** (about **3164.7 million bp**); the **10^6** is the source's own slip.*

- **DNA fingerprinting is a very quick way to compare the DNA sequences of any two individuals.** DNA fingerprinting involves **identifying differences in some specific regions in DNA sequence called as repetitive DNA**, because in these sequences, a **small stretch of DNA is repeated many times**.

These **repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation**. The **bulk DNA forms a major peak** and the **other small peaks are referred to as satellite DNA**.

Repetitive DNA	Satellite DNA
Sequences in which a **small stretch of DNA is repeated many times**. Identifying differences in these regions is what **DNA fingerprinting** does.	The **small peaks other than the major bulk-DNA peak** obtained when repetitive DNA is **separated from bulk genomic DNA during density gradient centrifugation**.
The **general class** -- any many-times-repeated stretch, making up a **large portion of the human genome**.	A **sub-class defined by its centrifugation behaviour**, and further classified **depending on base composition (A : T rich or G:C rich), length of segment, and number of repetitive units** into **micro-satellites, mini-satellites** etc.

These **sequences normally do not code for any proteins**, but they **form a large portion of human genome**. These **sequence show high degree of polymorphism and form the basis of DNA fingerprinting**.

Since **DNA from every tissue (such as blood, hair-follicle, skin, bone, saliva, sperm etc.), from an individual show the same degree of polymorphism**, they become **very useful identification tool in forensic applications**. Further, as the **polymorphisms are inheritable from parents to children**, **DNA fingerprinting is the basis of**

paternity testing, in case of disputes.

As **polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting**, it is essential that we understand **what DNA polymorphism means** in simple terms.

- **Polymorphism (variation at genetic level) arises due to mutations.** In simple terms, **if an inheritable mutation is observed in a population at high frequency, it is referred to as DNA polymorphism**. More precisely, **allelic sequence variation has traditionally been described as a DNA polymorphism if more than one variant (allele) at a locus occurs in human population with a frequency greater than 0.01**.

(Recall different kind of **mutations** and their effects that you have already studied in **Chapter 4**, and in the preceding sections in this chapter. Recall also the definition of **alleles** from Chapter 4.)

New mutations may arise in an individual either in somatic cells or in the germ cells (cells that generate gametes in sexually reproducing organisms). If a **germ cell mutation does not seriously impair individual's ability to have offspring who can transmit the mutation, it can spread to the other members of population** (through sexual reproduction).

The **probability of such variation to be observed in non-coding DNA sequence would be higher**, as **mutations in these sequences may not have any immediate effect/impact in an individual's reproductive ability**. These **mutations keep on accumulating generation after generation, and form one of the basis of variability/polymorphism**. There is a **variety of different types of polymorphisms ranging from single nucleotide change to very large scale changes**. For **evolution and speciation, such polymorphisms play very important role**, and you will study these in details at higher classes.

- The **technique of DNA Fingerprinting was initially developed by Alec Jeffreys**. He used a **satellite DNA as probe that shows very high degree of polymorphism**. It was called as **Variable Number of Tandem Repeats (VNTR)**.

The technique, as used earlier, **involved Southern blot hybridisation using radiolabelled VNTR as a probe**. It **involved the following steps**:

1. **(i) Isolation of DNA.**
2. **(ii) Digestion of DNA by restriction endonucleases.**
3. **(iii) Separation of DNA fragments by electrophoresis.**
4. **(iv) Transferring (blotting) of separated DNA fragments to synthetic membranes**, such as **nitrocellulose or nylon**.
5. **(v) Hybridisation using labelled VNTR probe.**
6. **(vi) Detection of hybridised DNA fragments by autoradiography.**

The **VNTR belongs to a class of satellite DNA referred to as mini-satellite**. A **small DNA sequence is arranged tandemly in many copy numbers**. The **copy number varies from chromosome to chromosome in an individual**. The **numbers of repeat show very high degree of polymorphism**. As a result the **size of VNTR varies in size from 0.1 to 20 kb**.

Consequently, after **hybridisation with VNTR probe, the autoradiogram gives many bands of differing sizes**. These **bands give a characteristic pattern for an individual DNA**. It **differs from individual to individual in a population except in the case of monozygotic (identical) twins**.

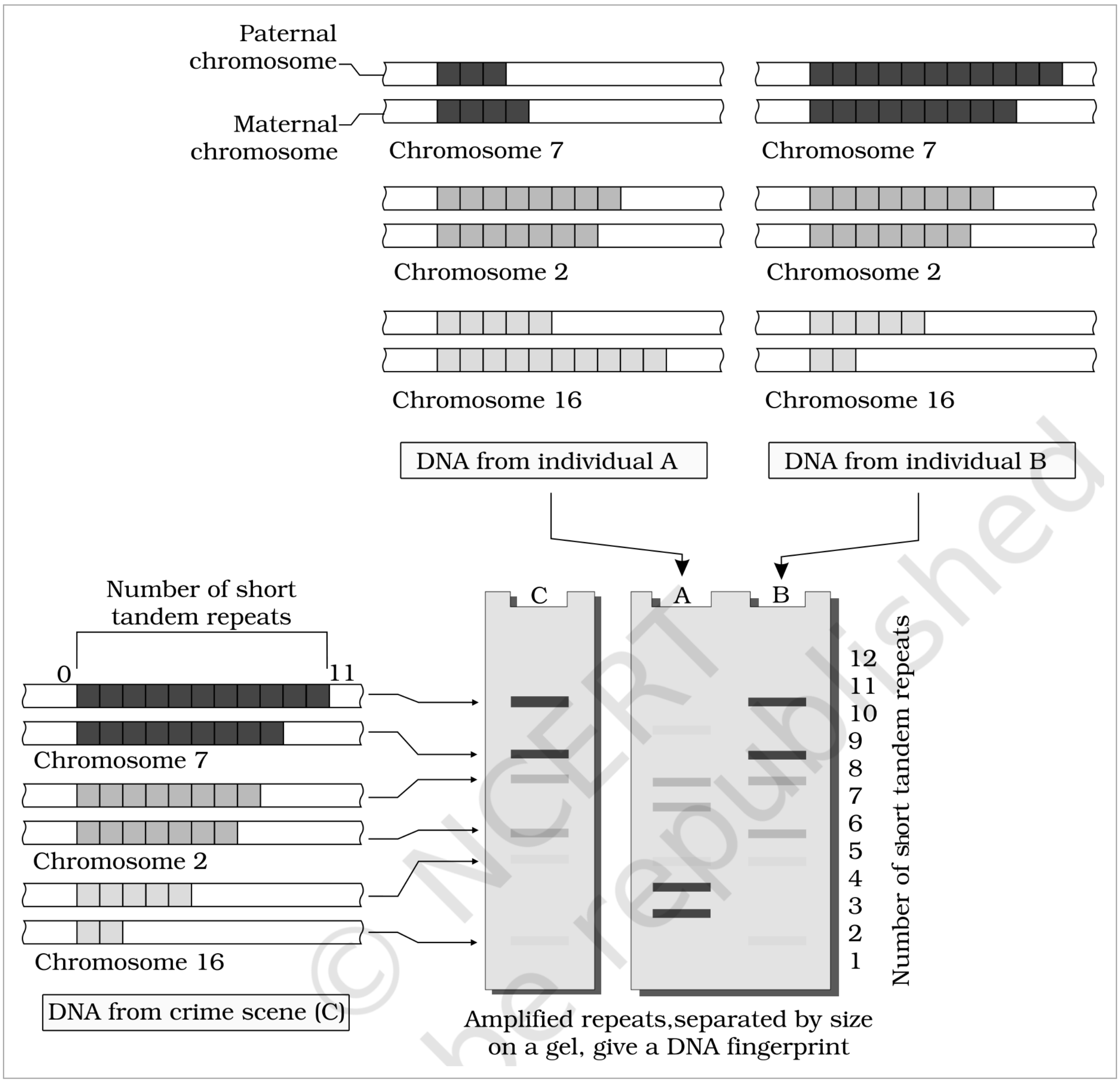


*Figure 5.16 Schematic representation of DNA fingerprinting. Repeats on the **Paternal chromosome** and the **Maternal chromosome** -- shown for **Chromosome 7**, **Chromosome 2** and **Chromosome 16** -- differ in the **Number of short tandem repeats**. **Amplified repeats, separated by size on a gel, give a DNA fingerprint**, so that **DNA from individual A**, **DNA from individual B** and **DNA from crime scene (C)** can be compared band for band.*

The **sensitivity of the technique has been increased by use of polymerase chain reaction (PCR** -- you will study about it in **Chapter 9**). Consequently, **DNA from a single cell is enough to perform DNA fingerprinting analysis**. **Currently, many different probes are used to generate DNA fingerprints.**

In addition to application in **forensic science**, it has much wider application, such as in **determining population and genetic diversities**. It has **immense applications in the field of forensic science, genetic biodiversity and evolutionary biology**.

☆ ***[MEMORY AID - not in NCERT]** The chain of logic: **99.9 per cent of human DNA is identical**, so you look only where it varies **--** the **repetitive / satellite DNA**, which is **non-coding** and therefore **free to accumulate mutations**, giving **high polymorphism**. **Alec Jeffreys**, probe = **VNTR**, a **mini-satellite**, size **0.1 to 20 kb**; the original readout was **Southern blot + autoradiography**, now made far more sensitive by **PCR -- a single cell is enough**. Two exam traps: **identical (monozygotic) twins share a fingerprint**, and **polymorphism counts as polymorphism only above a frequency of 0.01**.*

QR QUICK RECAP

This is the chapter's **SUMMARY**, rewritten denser: every sentence of the NCERT summary is accounted for here or has been folded into the body section it belongs to.

- **Nucleic acids are long polymers of nucleotides.** While **DNA stores genetic information, RNA mostly helps in transfer and expression of information**.
- Though **DNA and RNA both function as genetic material**, **DNA being chemically and structurally more stable is a better genetic material**. However, **RNA is the first to evolve and DNA was derived from RNA**.
- The **hallmark of the double stranded helical structure of DNA is the hydrogen bonding between the bases from opposite strands**. The rule is that **Adenine pairs with Thymine through two H-bonds, and Guanine with Cytosine through three H-bonds**. This **makes one strand complementary to the other**.
- The **DNA replicates semiconservatively**, the process being **guided by the complementary H-bonding**.
- A **segment of DNA that codes for RNA** may in a simplistic term be referred to as a **gene**. During **transcription** also, **one of the strands of DNA acts as a template to direct the synthesis of complementary RNA**.
- In **bacteria, the transcribed mRNA is functional, hence can directly be translated**. In **eukaryotes, the gene is split**: the **coding sequences, exons, are interrupted by non-coding sequences, introns**. **Introns are removed and exons are joined to produce functional RNA by splicing.**
- The **messenger RNA contains the base sequences that are read in a combination of three (to make triplet genetic code) to code for an amino acid**.
- The **genetic code is read again on the principle of complementarity by tRNA that acts as an adapter molecule**. There are **specific tRNAs for every amino acid**. The **tRNA binds to specific amino acid at one end and pairs through H-bonding with codes on mRNA through its anticodons**.
- The **site of translation (protein synthesis) is ribosomes**, which **bind to mRNA and provide platform for joining of amino acids**. **One of the rRNA acts as a catalyst for peptide bond formation, which is an example of RNA enzyme (ribozyme).**
- **Regulation of transcription is the primary step for regulation of gene expression.** In **bacteria, more than one gene is arranged together and regulated in units called as operons**. The **operon is regulated by the amount of lactose in the medium where the bacteria are grown**; therefore this **regulation can also be viewed as regulation of enzyme synthesis by its substrate**.
- **Human genome project was a mega project that aimed to sequence every base in human genome.** This **project has yielded much new information**, and **many new areas and avenues have opened up as a consequence of the project**.
- **DNA Fingerprinting is a technique to find out variations in individuals of a population at DNA level.** It **works on the principle of polymorphism in DNA sequences**.

EX TERMS USED IN THE EXERCISES

The end-of-chapter questions were scanned one by one against the body of this chapter. Most are answerable directly from the sections above; the few that lean on a convention or a step the source never spells out are closed **using only facts from this chapter**, in a **NOTE box beside the facts they depend on**. This table says where each answer lives.

Exercise	Where its answer lives
Q1 bases vs nucleosides	§5.1.1 -- a **nucleoside** is base + sugar (N-glycosidic linkage), and **adenosine, guanosine, cytidine, uridine** are named there. So **Cytidine** and **Guanosine** are nucleosides; **Adenine, Thymine, Uracil, Cytosine** are bases.
Q2 per cent of adenine	§5.1.1 -- **worked NOTE box** beside Chargaff's rule and the A-T / G-C pairing rule.
Q3, Q4 complementary strand and mRNA of a given 28-mer	§5.5.1 -- **worked NOTE box** beside the template / coding strand definitions, doing the given sequence once for the complementary strand and once for the mRNA.
Q5 which property suggested semi-conservative replication	§5.1.1 and §5.4 -- **complementary base pairing**, and Watson and Crick's own statement that it **suggests a mechanism for copying**.
Q6 types of nucleic acid polymerases	§5.4.2 -- **NOTE box** laying the classes out as a template-versus-product grid.
Q7 how Hershey and Chase told DNA from protein	§5.2.1 -- **radioactive phosphorus** labels DNA only (protein has no P), **radioactive sulfur** labels protein only (DNA has no S).

Exercise	Where its answer lives
Q8 (a) repetitive vs satellite DNA, (b) mRNA vs tRNA, (c) template vs coding strand	(a) §5.10 -- the **two-column table** in that section. (b) §5.5.3 (mRNA is the template) and §5.6.2 (tRNA is the adapter). (c) §5.5.1 -- **template strand is 3'-to-5' and is copied**; **coding strand is 5'-to-3' and codes for nothing**.
Q9 two essential roles of the ribosome	§5.7 -- the **two binding sites in the large subunit** that hold successive amino acids close enough to react, and the **23S rRNA ribozyme** that catalyses the peptide bond.
Q10 why the lac operon shuts down again after lactose is added	§5.8.1 -- **NOTE box** after negative regulation: the induced beta-galactosidase **consumes the inducer**, so the repressor re-binds the operator.
Q11 function of promoter, tRNA, exons	§5.5.1 (promoter -- binding site for RNA polymerase, defines template and coding strands), §5.6.2 (tRNA -- adapter, reads codon and carries amino acid), §5.5.2 (exons -- expressed sequences that appear in mature RNA).
Q12 why HGP is called a mega project	§5.9 -- the **mega project table**: 3 x 10^9 bp at US $3 per bp, about 9 billion dollars, 3300 books to print one cell's sequence, 13 years, high-speed computation.
Q13 what DNA fingerprinting is, and its applications	§5.10 -- definition, **Alec Jeffreys** and the **VNTR** probe, the **six steps**, and the applications (**forensic science, paternity testing, population and genetic diversities, genetic biodiversity, evolutionary biology**).
Q14 (a) transcription, (b) polymorphism, (c) translation, (d) bioinformatics	(a) §5.5 opener, (b) §5.10 (variation arising from mutation, above a frequency of 0.01), (c) §5.7 opener, (d) §5.9 -- **NOTE box** on what bioinformatics does.
In-body exercises beneath Table 5.1	§5.6 -- both are **worked in full** in the NOTE box under the codon checker-board, including why the reverse direction is ambiguous.